



Savitribai Phule Pune University
(Formerly University of Pune)

T.Y.B.Sc.(CA)/ T.Y.B.C.A. – Semester-V

(Faculty of Science and Technology)

(For Colleges Affiliated to Savitribai Phule Pune University)

**Choice Based Credit System (CBCS) Syllabus Under
National Education Policy (NEP)**

To be implemented from Academic Year 2026-2027

Title of the Course: T.Y.B.Sc.(CA)/T.Y.B.C.A.



**Savitribai Phule Pune University (Formerly
University of Pune)**

T.Y.B.Sc.(CA) /T.Y.B.C.A. Semester-V

WORKBOOK

Course Code: CA-313-MJP

Course Title:

Lab Course on CA-312-MJ Cloud Computing

Student Name			
College Name			
Roll No.		Division	
Exam Seat No.		Academic Year	

From the Chairman's Desk

It gives me great pleasure to present this workbook prepared by the Board of Studies in Computer Applications. The workbook has been prepared with the objectives of bringing uniformity in the implementation of lab assignments across all affiliated colleges, acting as a ready reference for both fast and slow learners, and facilitating continuous assessment using clearly defined rubrics.

The workbook provides, for each of the assignments, the aims, prerequisites, related theoretical concepts with suitable examples wherever necessary, guidelines for the faculty/lab administrator, instructions for the students to perform assignments, and a set of exercises divided into three sets.

I am thankful to the Chairman of this course and the entire team of editors. I am also thankful to the reviewers and members of BOS, **Dr. A.B. Nimbalkar** and **Dr. Razzak Sayyad**. I thank all members of BOS and everyone who has contributed directly or indirectly to the preparation of the workbook.

Constructive criticism is welcome and should be communicated to the Chairman of the Course and overall coordinator, **Dr. A. B. Nimbalkar**. Affiliated colleges are requested to collect feedback from the students for further improvements.

I am thankful to Hon. Vice Chancellor of Savitribai Phule Pune University **Prof. Dr. Suresh Gosavi**, Pro Vice Chancellor **Dr. Parag Kalkar**, and the Dean of Faculty of Science and Technology **Prof. Dr. Pramod D. Patil** for their support and guidance.

Prof. Dr. S. S. Sane

Chairman, BOS in Computer Applications

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5	Containerization using Docker	57

Assignment Completion Sheet

CA-312-MJ Cloud Computing Assignments Internal Evaluation			
Sr. No.	Assignment Name	Marks (Out of 5)	Instructor Sign
1	Working and implementation of Software as Service(SAAS)		
2	Understanding Virtualization using Virtual box		
3	Introduction to Google Cloud Console & Cloud shell Editor		
4	Introduction to cloud Platform AWS & Microsoft Azure		
5	Containerization using Docker		

Batch-Incharge Sign

Certificate

This is to certify that **Mr./Ms.**_____

Exam Seat No. _____ have successfully completed and submitted the coursework for lab course **CA- 312-MJ** on **Cloud Computing, T.Y.B.Sc.(C.A.)/ TYBCA – Sem. V** prescribed by Savitribai Phule Pune University during Academic year-_____and has scored marks_____out of 15 Marks.

Batch-Incharge

Head of the Department./Coordinator

Internal examiner

External examiner

1. About the work book:

This LAB book / workbook is intended to be used by T.Y.B.C.A. / T.Y.B.Sc. (Computer application) students for the **Cloud Computing** assignments in **Semester–V**. This workbook is designed by considering all the practical concepts / topics mentioned in syllabus. The lab book is to be used as a hands-on resource, reference and record of assignment submission and completion by the student. The lab book contains the set of assignments which the student must complete as a part of this course.

2. The objectives of this LAB-Book are:

- Defining the scope of the course.
- To bring uniformity in the practical conduction and implementation in all colleges affiliated to SPPU.
- To have continuous assessment of the course and students.
- Providing ready reference for the students during practical implementation.
- Provide more options to students so that they can have good practice before facing the examination.
- Catering to the demand of slow and fast learners and accordingly providing the practice assignments to them.

3. Instructions to the students:

- Students are expected to carry this book every time they come to the lab for computer science practical.
- Students should prepare oneself beforehand for the assignment by reading the relevant material.
- Instructor will specify which problems to solve in the lab during the allotted slot and student should complete them and get verified by the instructor.
- However student should spend additional hours in Lab and at home to cover as many problems as possible given in this work book.

4. Submission of Assignments:

- The Cloud Computing assignments are to be submitted by the student in the form of a journal containing each assignment sheet.
- Each assignment includes the Assignment Title, Problem statement, Date of submission, Assessment date, Assessment Marks or Grade and Batch In charge's sign.

5. Assessment:

- Continuous assessment of laboratory work is to be done based on overall performance and lab assignments performance of student.
- Each lab assignment assessment will be assigned grade / marks based on parameters with appropriate weightage.
- Suggested parameters for overall assessment as well as each lab assignment assessment includes: timely completion, performance, innovation, efficient codes and good programming practices.
- Students will be assessed for each exercise on a scale from 0 to 5.

Not done	0	Needs improvement	3
Incomplete	1	Complete	4
Late Complete	2	Well Done	5

Assignment No.1

Title: Working and implementation of Software as a Service

Prerequisite:

- Basic knowledge of Cloud Computing.
- A valid Google account is required.
- A stable internet connection is necessary.

Guidelines for Teachers/Instructors:

- Prepare workspace before assignment
- Clearly define assignment objectives
- Select appropriate SaaS tools
- Set permissions and access control carefully
- Shares file properly

Instructions for Students:

- Students must have a valid Google account.
- Carefully read instructions provided by the teacher.
- Understand the objective of assignment clearly.
- Solve Set A, B and C assigned by instructor in allocated slots only.

1. Introduction:

What is Cloud Computing?

Cloud computing is a technology that allows users to access computing resources such as servers, storage, databases, networking, and software over the internet. Instead of purchasing own physical hardware or software, users can access services on-demand.

Cloud computing is broadly divided into three service models:

- Infrastructure as a Service (IaaS)
- Platform as a Service (PaaS)
- Software as a Service (SaaS)

Characteristics of Cloud Computing

- **On-Demand Self-Service:** You can provision resources (like server time or network storage) automatically without needing a human at the service provider.
- **Broad Network Access:** Services are available over the network and accessed through standard mechanisms (mobile phones, tablets, laptops).

- **Resource Pooling:** The provider's resources are pooled to serve multiple consumers using a multi-tenant model.
- **Rapid Elasticity:** You can scale up or scale down instantly based on demand. If your website goes viral, the cloud expands; when traffic drops, it shrinks.
- **Measured Service:** You are billed based on a "pay-as-you-go" model, much like a water or gas meter.

Deployment Models

Where exactly is the cloud located, and who has access to it?

- **Public Cloud:** Owned and operated by third-party providers. You share hardware with other "tenants." (e.g., AWS, Azure).
- **Private Cloud:** Cloud resources used exclusively by one business or organization. It can be physically located at your on-site data centre or hosted by a provider.
- **Hybrid Cloud:** A mix of both. You might keep sensitive financial data on a private cloud but run your public-facing web app on a public cloud.
- **Multi-Cloud:** Using services from multiple different public cloud providers (e.g., using AWS for storage and Google for AI tools) to avoid "vendor lock-in."

Service Models: The "As-A-Service" Stack

Cloud computing is generally divided into three layers, depending on how much control you want versus how much management you want to offload.

Model	Full Name	What you manage	Example
IaaS	Infrastructure as a Service	Operating systems, apps, and data.	Amazon EC2, Azure VMs
PaaS	Platform as a Service	Only your applications and data.	Heroku, Google App Engine
SaaS	Software as a Service	Nothing to manage you just use the software.	Gmail, Salesforce, Slack

Introduction to Software as a Service (SaaS)

Software as a Service (SaaS) is a cloud-based software delivery model where applications are hosted by a service provider and made available to users over the internet.

Instead of installing software on individual machines, users access it via a web browser.

Examples of SaaS Applications:

- Google Docs
- Microsoft 365
- Dropbox
- Salesforce

Characteristics of SaaS

1. Web-Based Access

Users access applications through browsers.

2. Centralized Hosting

Software is hosted on remote servers.

3. Subscription-Based Pricing

Users pay monthly or yearly.

4. Automatic Updates

Updates are handled by the provider.

5. Multi-Tenant Architecture

Multiple users share the same infrastructure.

Working of SaaS (Step-by-Step)

Step 1: User Access

- The user opens a web browser.
- Logs into the SaaS application using credentials.

Step 2: Request Transmission

- The browser sends a request to the cloud server.
- This request travels via the internet.

Step 3: Authentication & Authorization

- The server verifies the user identity.
- Access rights are checked.

Step 4: Application Processing

- The SaaS application processes user requests.
- Logic is executed on the cloud server.

Step 5: Data Storage & Retrieval

- Data is stored in cloud databases.
- Required data is fetched when requested.

Step 6: Response to User

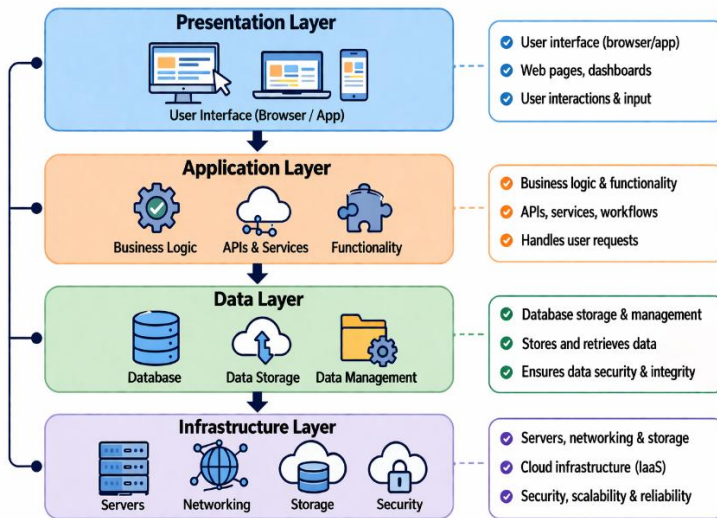
- The server sends the processed result back.
- The user sees output on the browser.

Architecture of SaaS

SaaS architecture mainly consists of:

- 1. Presentation Layer:** User interface (browser/app)
- 2. Application Layer:** Business logic and functionality
- 3. Data Layer:** Database storage and management
- 4. Infrastructure Layer:** Servers, networking, and storage

Architecture of SaaS (Software as a Service)



SaaS get started

To start SaaS application, we need stable internet connection and google account.

Here is a **list of Google applications that use the SaaS (Software as a Service) model**, similar to Google Drive and Google Sheets:

Google Applications Using SaaS

Google Applications	Uses
Google Drive	Cloud storage and file sharing
Google Docs	Online document editor
Google Sheets	Online spreadsheet tool
Google Slides	Presentation creation
Google Forms	Surveys and quizzes
Gmail	Email service
Google Meet	Video meetings
Google Chat	Team messaging
Google Calendar	Scheduling and planning
Google Keep	Notes and reminders
Google Sites	Create websites
Google Colab	Python coding in browser

Let us see step-by-step process to learn google applications use as Software-as-a-Service.

1. Google Drive:

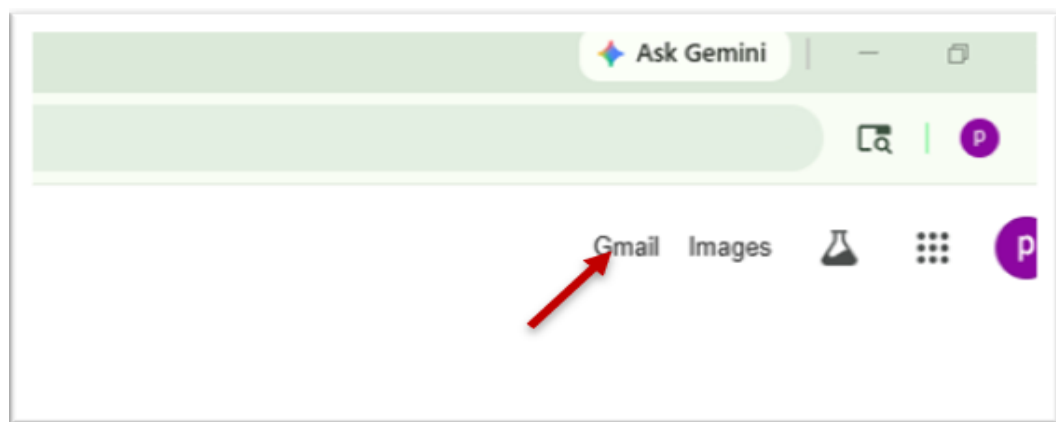
Google Drive is a cloud storage service provided by Google. Its main purpose is to help you store, access, and manage your files online instead of only on your device.

Here are its key purposes:

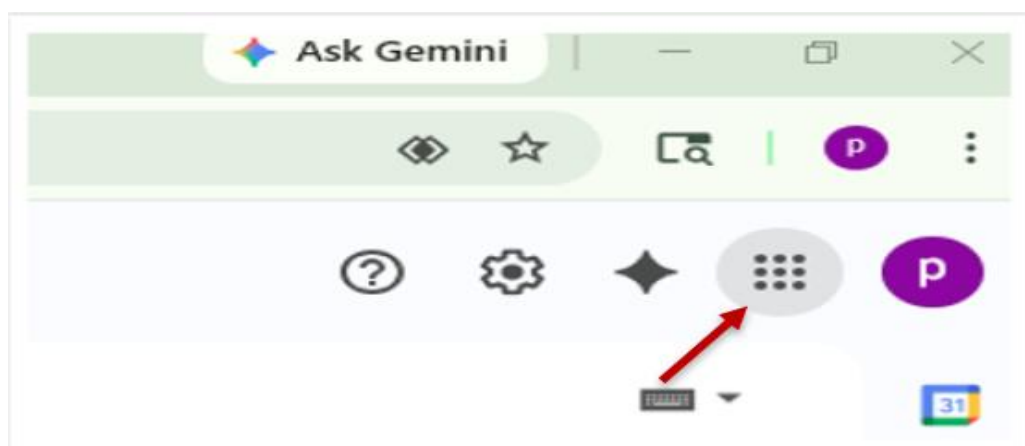
- **Store files online:** You can save documents, photos, videos, and other files securely in the cloud.
- **Access from anywhere:** You can open your files on any device (phone, tablet, or computer) with internet access.
- **Backup data:** It acts as a backup so your files are safe even if your device is lost or damaged.
- **Share files easily:** You can share files or folders with others using a link or by inviting them.
- **Collaborate in real time:** Multiple people can work on the same document (Google Docs, Sheets, Slides) at the same time.
- **Organize files:** You can create folders, rename files, and keep everything neatly arranged.

➤ How to use google drive:

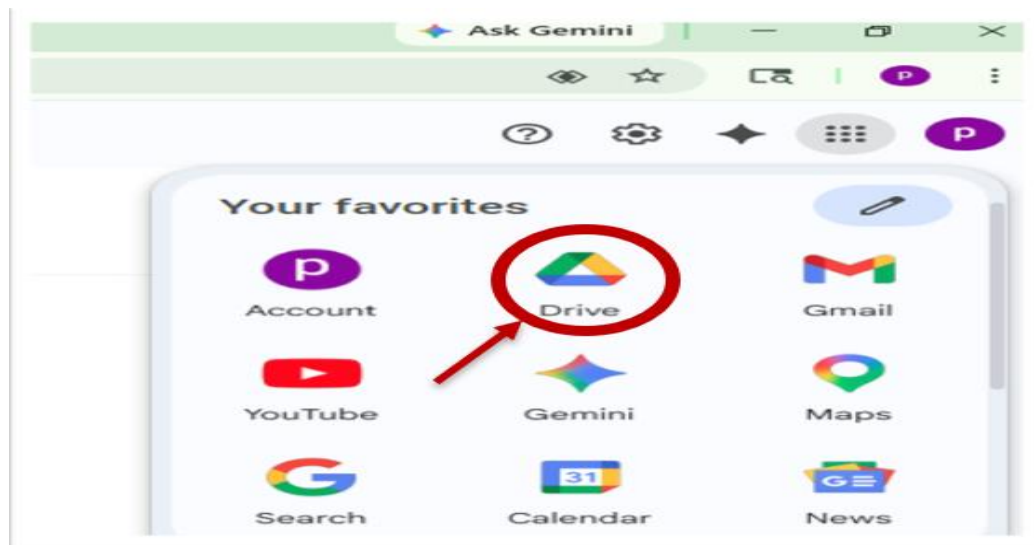
1. First Sign in your google account:



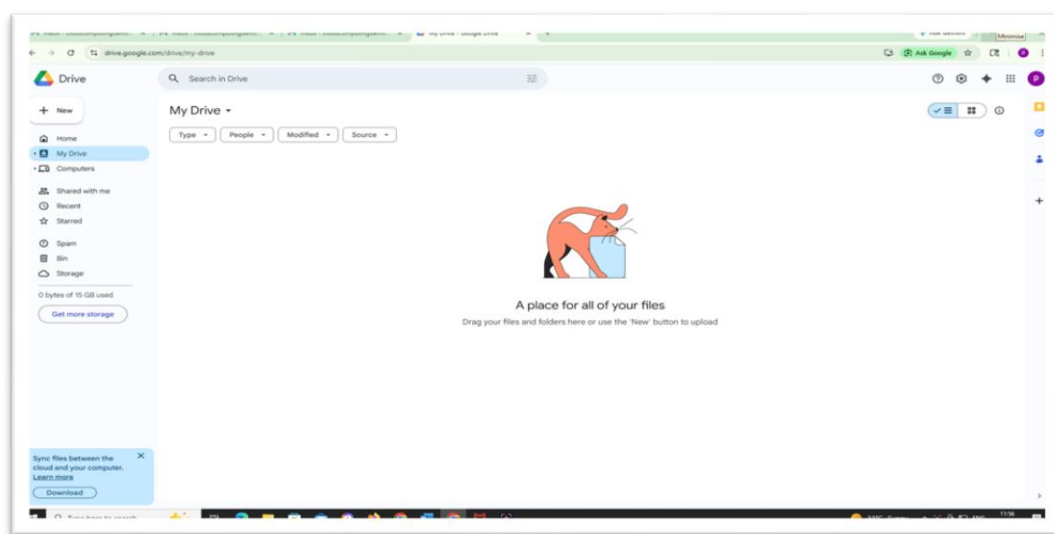
2. After sign in click on google apps (i.e. 9 dots) shown on our right-hand side.



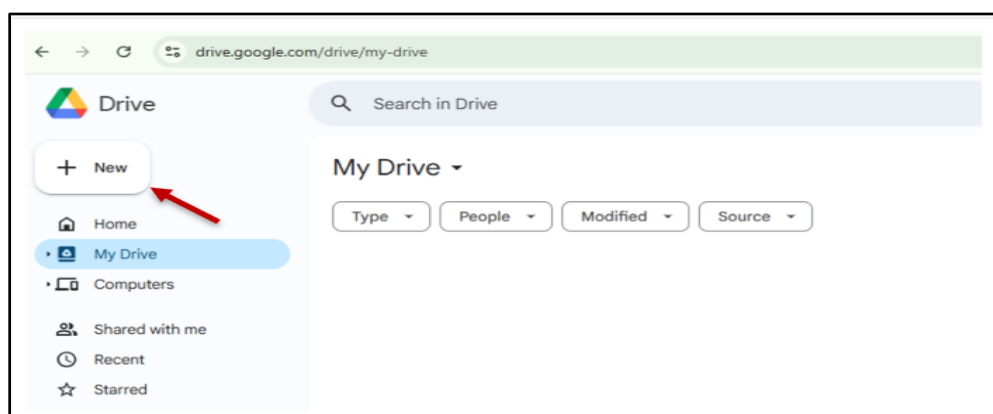
3. Select drive option.



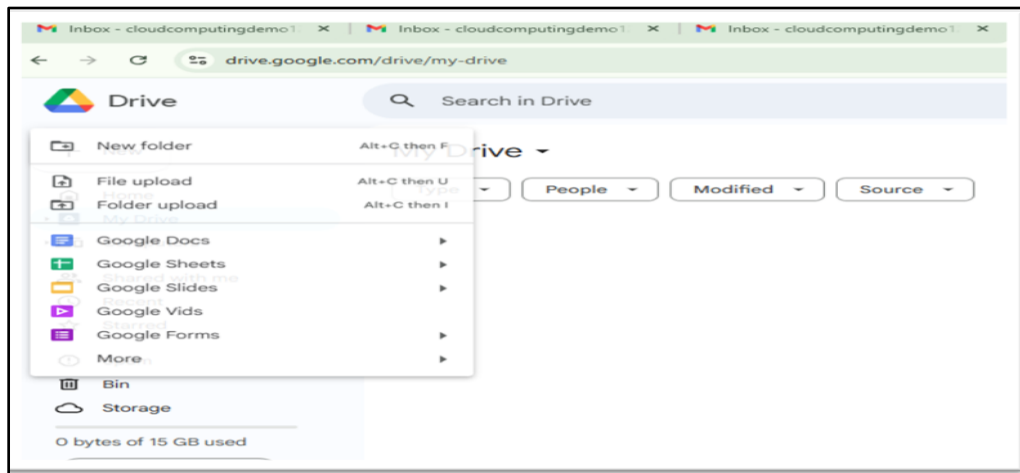
4. After clicking on drive the next window will be open:



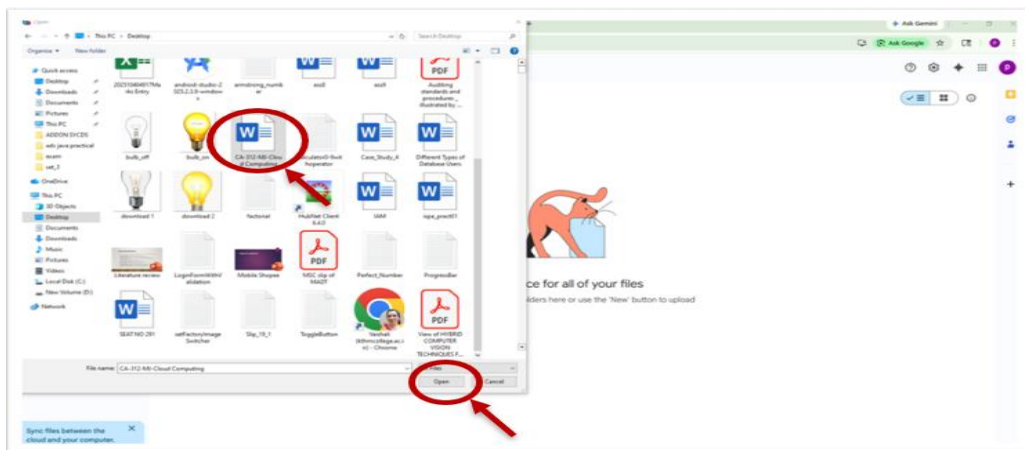
5. Select + **New** option in drive that are shown in our left-hand side.



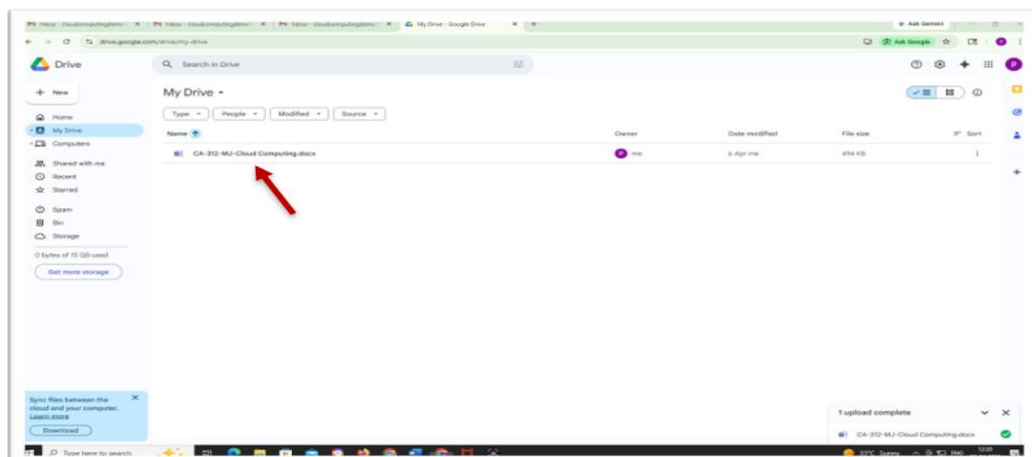
6. After click on + **New** we see the options like new folder, file upload, folder upload etc.



7. Select any one option that u want to be perform action. Suppose I select file upload option the it will go to browse the file from your device. You can select any file and click on open button i.e. in bottom of selection tab.



8. After click on open button file will be uploaded successfully.



9. In the same way you can create new folder, upload folder to Store files online and access from anywhere.

2. Google Sheet:

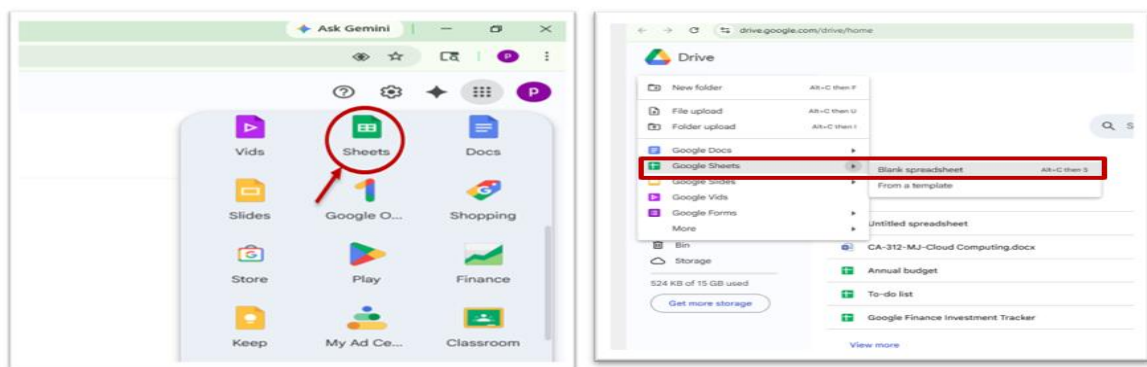
The purpose of Google Sheets for people is to help individuals and groups manage data, perform calculations, and work together efficiently in everyday tasks.

Key purposes:

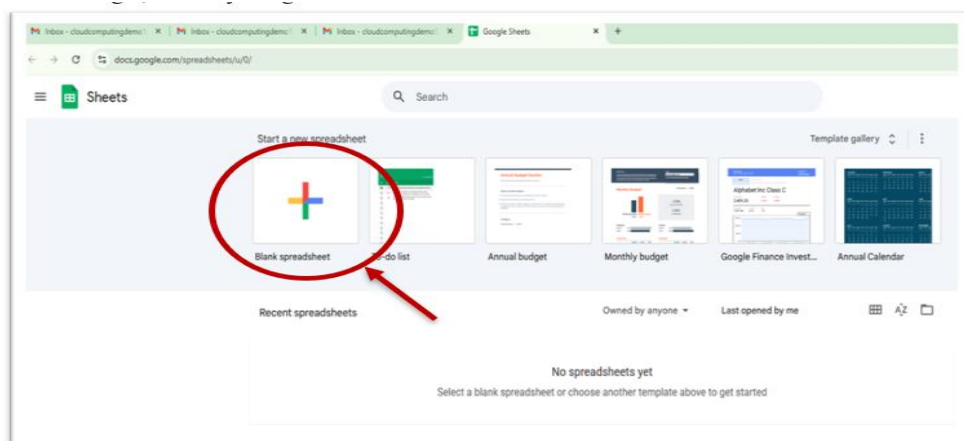
- **Personal use:** Track expenses, budgets, daily tasks, study schedules, or fitness records.
- **Students:** Manage marks, assignments, attendance, and project data in an organized way.
- **Office workers / professionals:** Handle reports, sales data, inventory, employee records, and financial analysis.
- **Collaboration:** Multiple people (team members, classmates, etc.) can work on the same sheet at the same time.
- **Decision making:** Use charts and formulas to analyze data and make better decisions.
- **Easy sharing & access:** Share sheets with others and access them anytime, anywhere.

How to use google sheets:

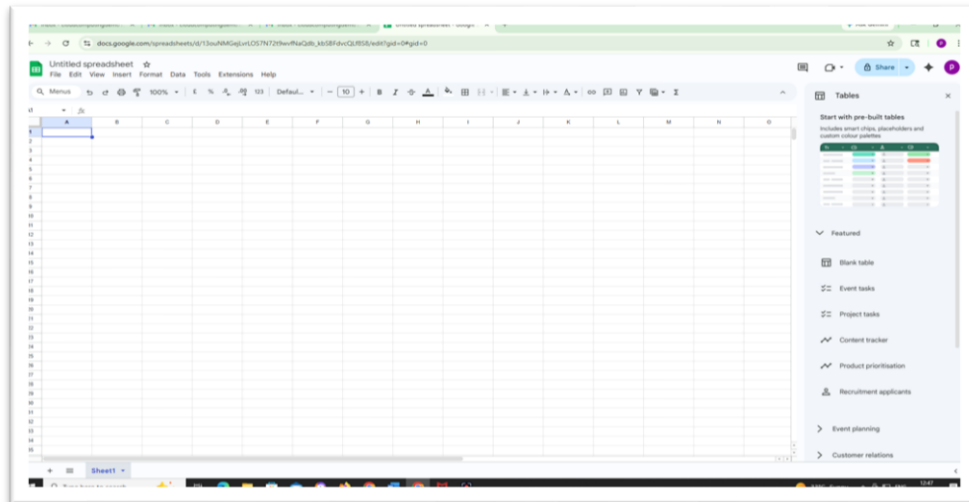
1. There are two ways to create google sheets. 1. Select **Sheets** option from google apps (i.e. 9 dots) shown on our right-hand side. 2. Click on “+ New” option in drive, Select **Google Sheets** → **Blank Spreadsheet**.



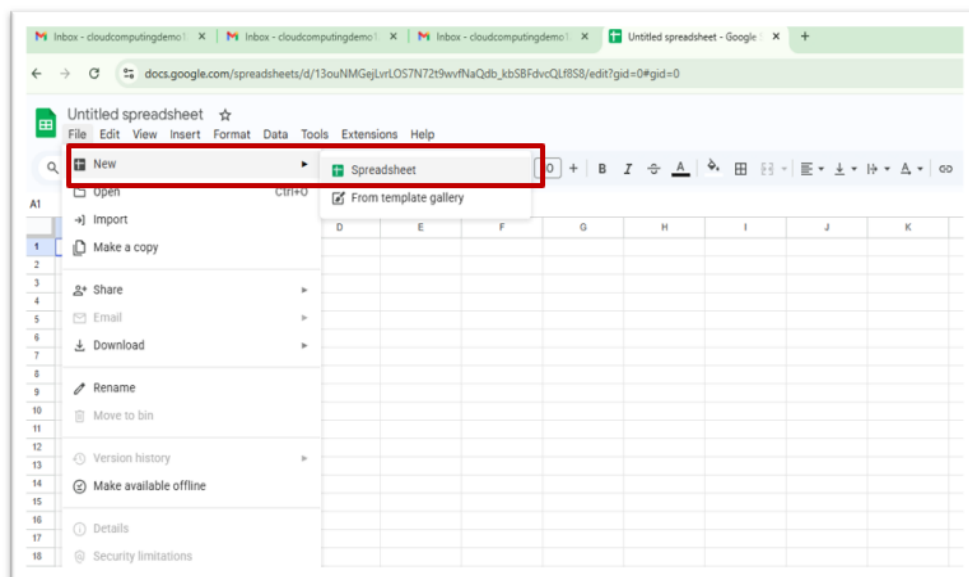
2. After click on google sheets, a new tab will open with various spreadsheet like blank spreadsheet, To-do list, Annual budget, Monthly budget etc.



3. Now select **Blank Spreadsheet** option. The screen will open like this.



4. After click on file menu select New → Spreadsheet.



5. After selecting spreadsheet, you can add various records.

	A	B	C	D	E	F	G	H
1	Roll Number	Student Name	Class	Address	Phone Number	Result		
2	1	XYZ	FYBCA	Nashik	2536147891	Pass		
3	2	ABC	SYBCA	Malegaon	5268471665	Pass		
4	3	PQR	SYBCA	Nagar	8564778984	Fail		
5								
6								
7								

6. There is no need to save the sheet it will automatically save in your drive. You can also perform various

operation like create chart, apply filter to data etc.

3. Google Docs:

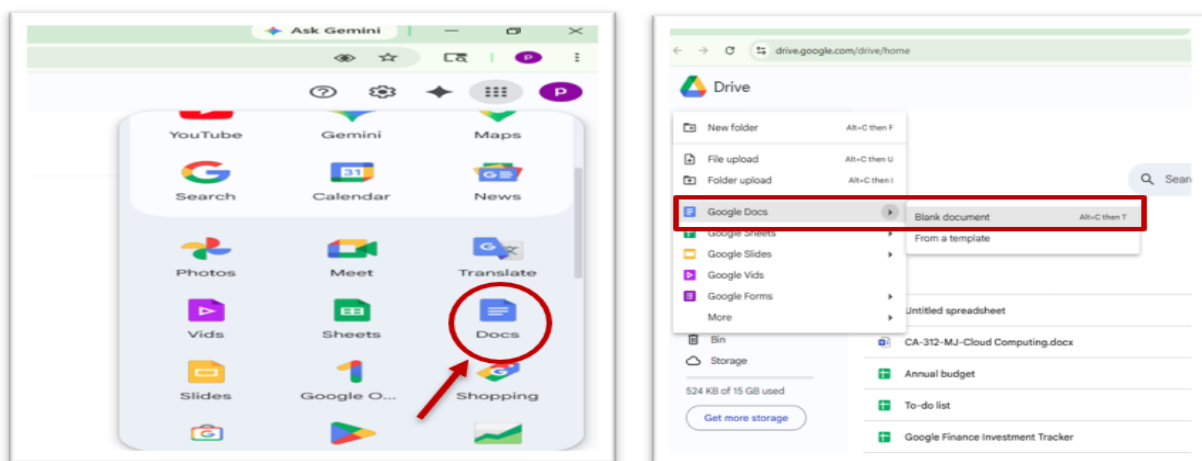
The purpose of Google Docs is to create, edit, format, and share documents online.

Key purposes:

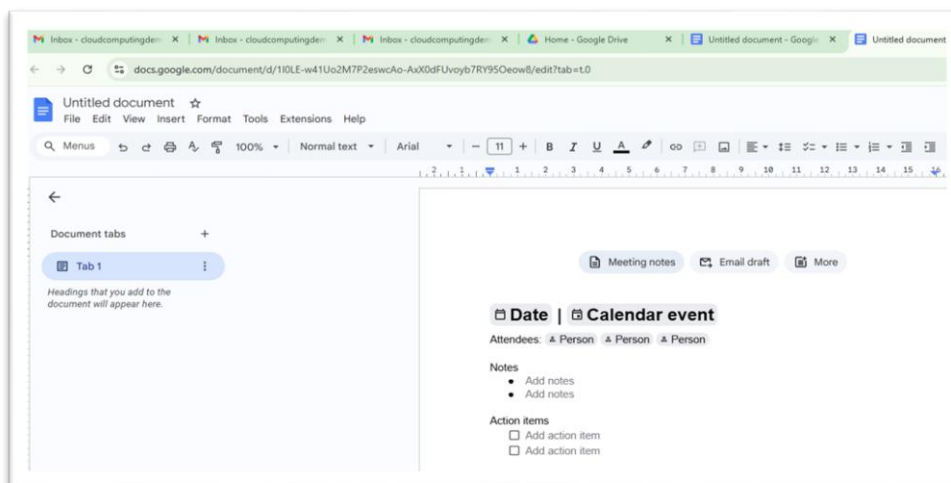
- **Create documents:** Write essays, reports, letters, notes, and more.
- **Edit and format content:** Use tools like bold, headings, fonts, alignment, bullet points, etc.
- **Real-time collaboration:** Multiple users can work on the same document at the same time.
- **Auto-saving:** Your work is saved automatically without needing to click “Save”.
- **Easy sharing:** Share documents with others using a link or email.
- **Access anywhere:** Open your documents from any device with internet access.

How to use google docs:

1. There are two ways to create google docs. 1. Select **Docs** option from google apps (i.e. 9 dots) shown on our right-hand side. 2. Click on “+ **New**” option in drive, Select **Google Docs** → **Blank documents**.

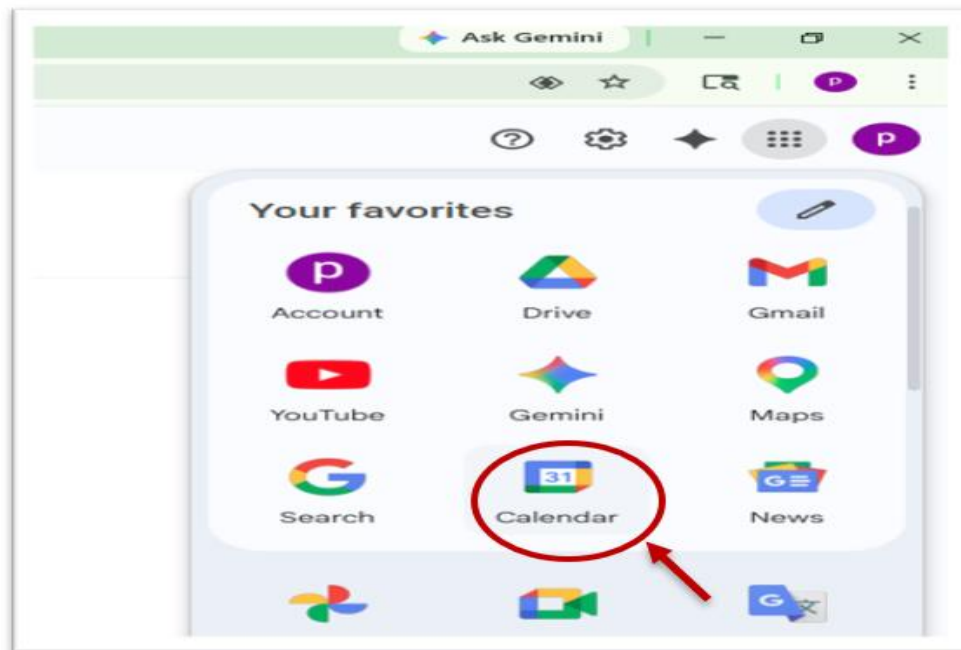


2. After click on the Blank documents the window will open like below. Here you can add meeting notes, Email Draft or more

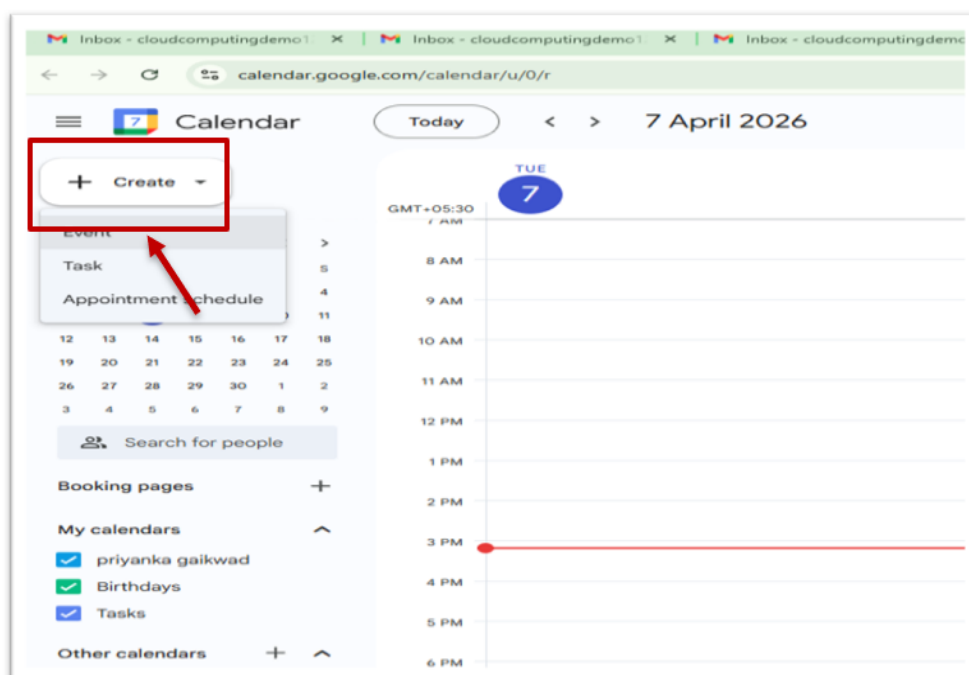


3. To add meeting notes first of all you should create a **calendar** event using **Calendar** option from google

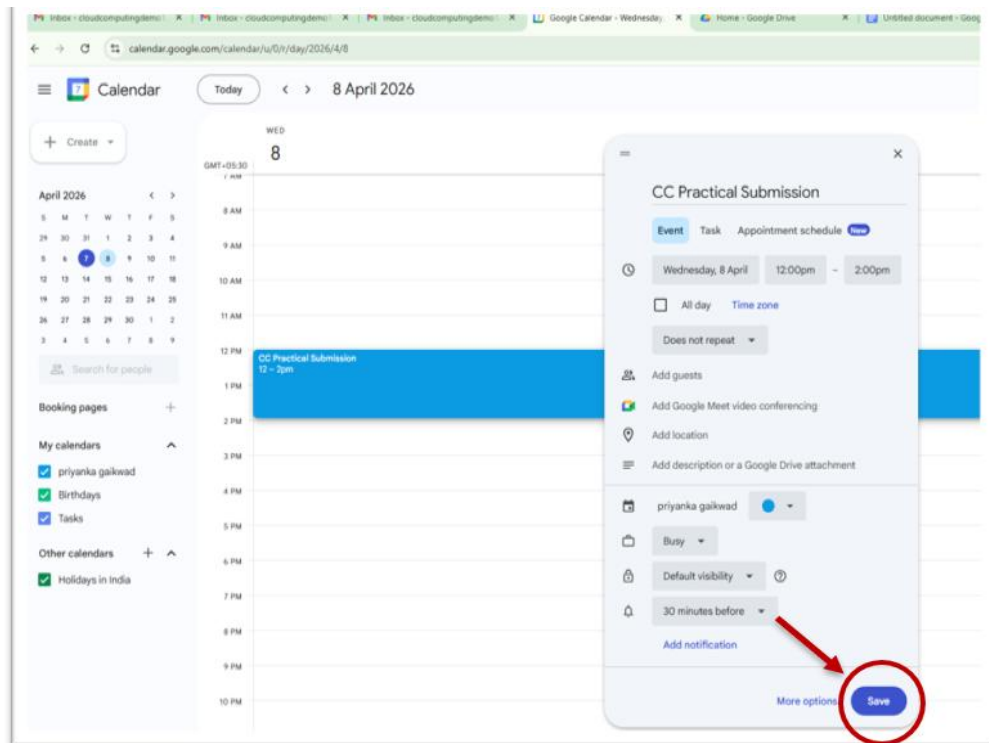
apps (i.e. 9 dots) shown on our right-hand side.



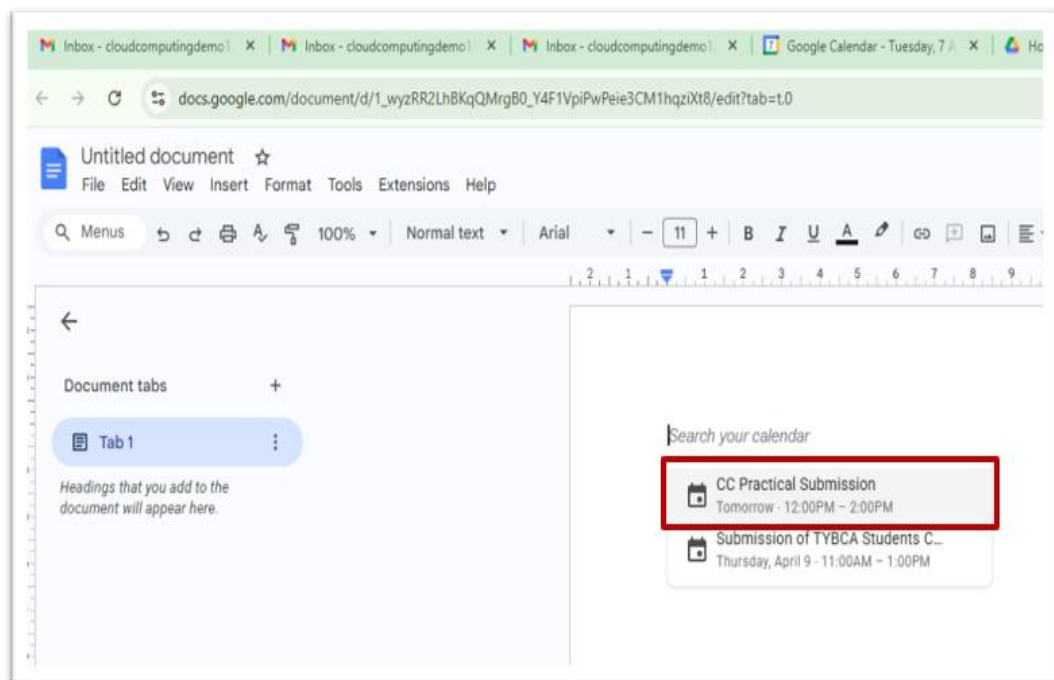
4. After selecting Calendar Click on + **Create** button and select **Event** option.



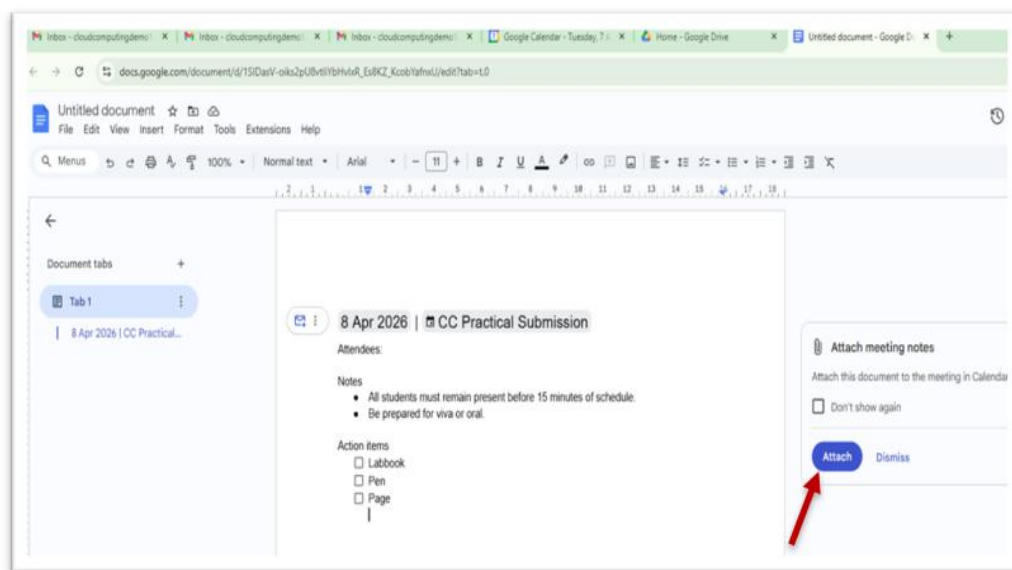
5. Add event title, set date and time of event, and also set the notification alarm. Click on **Save** button.



6. After saving event, open **google docs** and click on **Meeting Notes**, the event will be shown. Select any one event where you have to add some notes. Here I am selecting **CC Practical Submission** event.



7. You can add the notes on event like as shown below, if you want to attach these notes with calendar then you can attach it using **Attach** button.



8. In this way you can also create **Email Draft** using Google Docs and send email.

4. Google Forms:

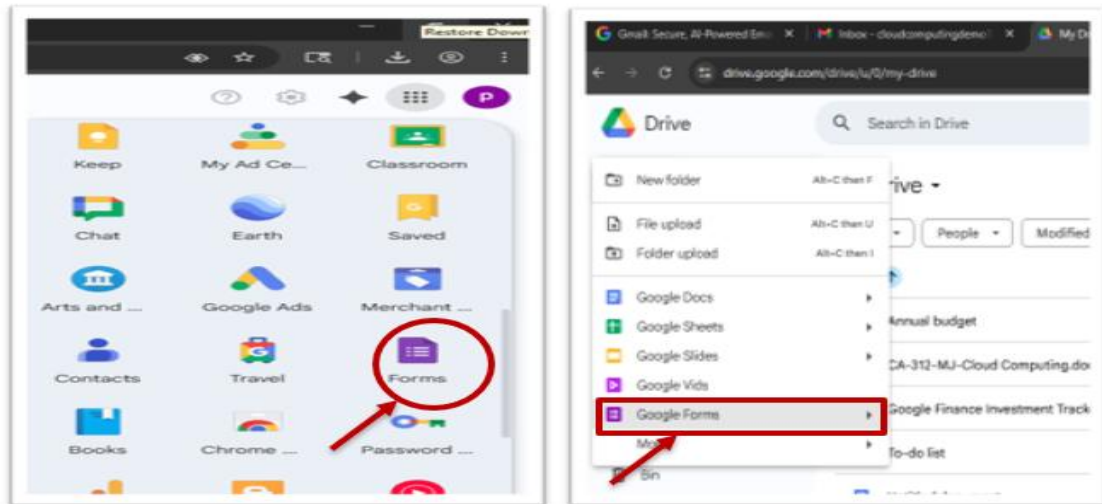
The **purpose of Google Forms** is to **collect, organize, and analyse information from people easily and efficiently**. It is widely used by individuals, students, teachers, and organizations.

Key Purposes:

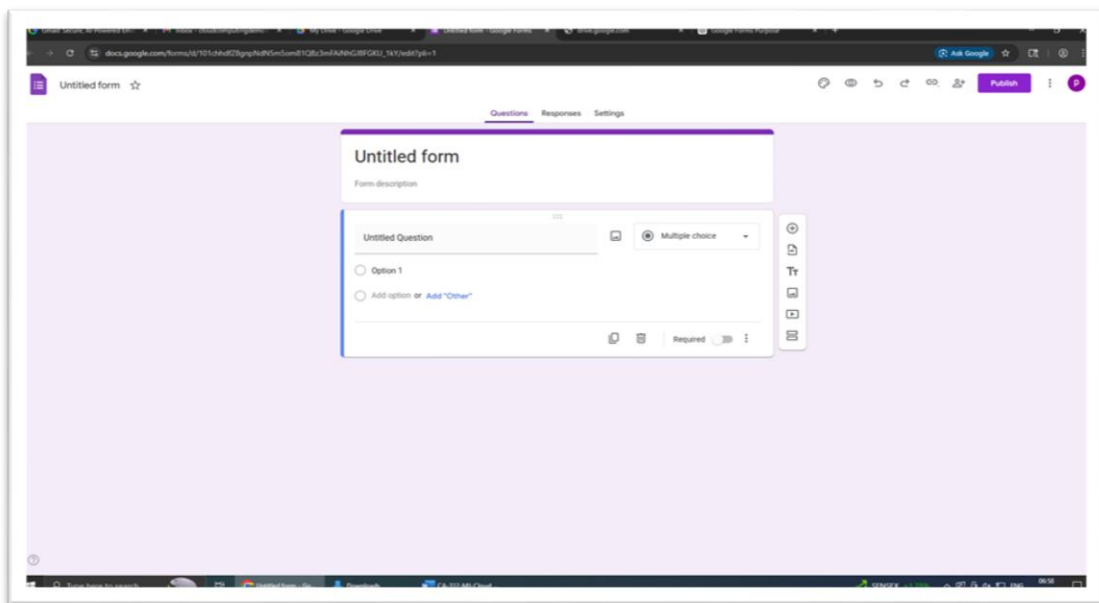
- **Data Collection:** Gather responses from people through surveys, questionnaires, or forms. Example: feedback forms, registration forms.
- **Online Surveys:** Create surveys for research, opinions, or decision-making. Useful in academic and business environments.
- **Quizzes & Exams:** Teachers can create quizzes with automatic grading. Saves time and provides instant results.
- **Event Registration:** Collect participant details for events, workshops, or webinars.
- **Feedback & Reviews:** Get feedback from customers, students, or employees to improve services.
- **Collaboration:** Multiple people can create and edit forms together in real-time.
- **Data Analysis:** Responses are automatically stored and can be linked to Google Sheets for easy analysis using charts and graphs.

How to use google forms:

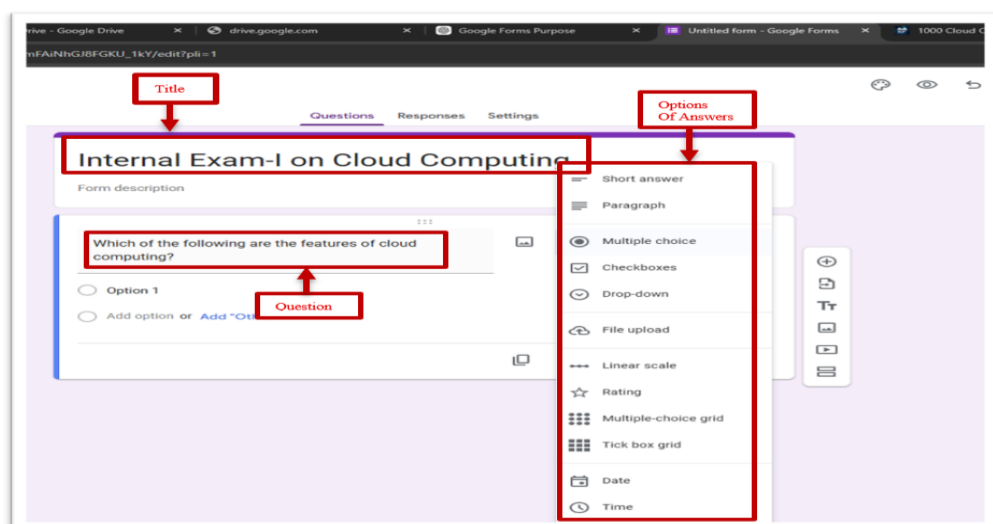
1. There are two ways to create google forms. 1. Select **Forms** option from google apps (i.e. 9 dots) shown on our right-hand side. 2. Click on “+ New” option in drive, Select **Google Forms**.



2. After click on google forms the window will appear like below:



3. Give proper **title** to google form and add questions having various options of answers like, short answer, paragraph, multiple choice, checkbox etc. is shown below:



4. Here, I select Multiple choice then add options for MCQ.

Internal Exam-I on Cloud Computing

Form description

Which of the following are the features of cloud computing?

Multiple choice

Security
Availability
Large Network Access
All of the mentioned
Add option or Add "Other"

Answer key (0 points)

Required

5. If **answer key** option at the bottom is not visible then go to setting and select **Make this a quiz** option ON otherwise forget it.

Settings

Make this a quiz
Assign point values, set answers and automatically provide feedback

RELEASE MARKS

Immediately after each submission
Later, after manual review
Turns on responses → collect email addresses

RESPONDENT SETTINGS

Missed questions
Respondents can see which questions were answered incorrectly

Correct answers
Respondents can see correct answers after grades are released

Point values
Respondents can see total points and points received for each question

GLOBAL QUIZ DEFAULTS

Default question point value
Point values for every new question

0 points

6. Click on **Answer Key**, select correct option and give the **points** to correct answer after that click on **Done** Button.

Internal Exam-I on Cloud Computing

Form description

Choose correct answers:

Which of the following are the features of cloud computing?

Security
Availability
Large Network Access
All of the mentioned
Add answer feedback

1 points

Done

7.If the question is compulsory to answer, then select **Required** ON.

The screenshot shows a form titled "Internal Exam-I on Cloud Computing" with tabs for "Questions", "Responses", and "Settings". The "Questions" tab is active. The form contains a question: "Which of the following are the features of cloud computing?" with a "Multiple choice" dropdown. The question has five options: "Security", "Availability", "Large Network Access", "All of the mentioned", and "Add option or Add 'Other'". The "All of the mentioned" option is selected with a green checkmark. At the bottom of the question, there is a "Required" toggle switch which is turned ON (indicated by a red circle and an arrow).

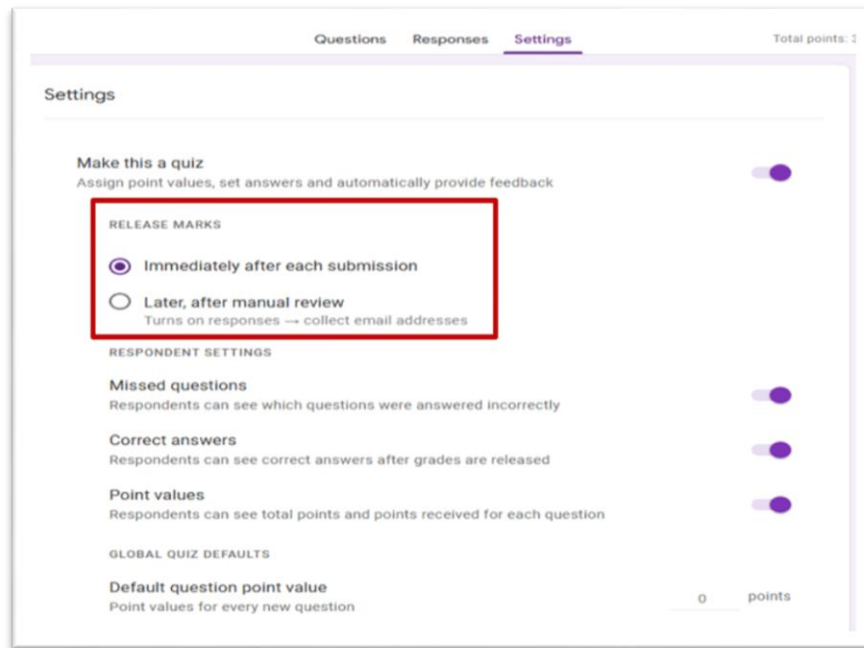
8. You can add new question using + sign shown in our right-hand side of form.

The screenshot shows the same form as before, but with a red circle and an arrow pointing to a "+" sign in the right-hand side of the form, indicating where to click to add a new question.

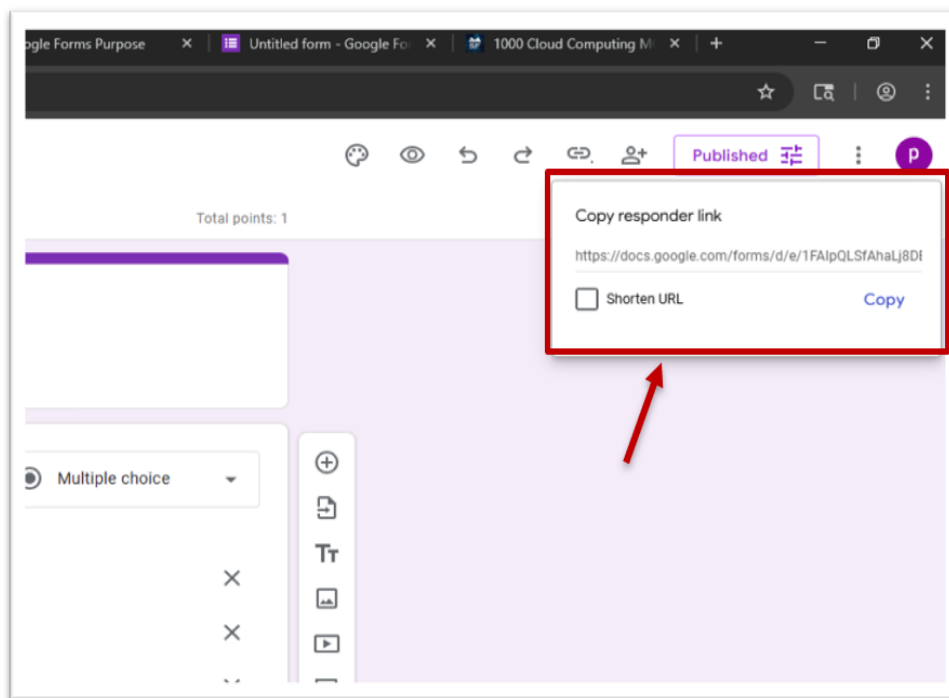
9. Added 2nd question with correct answer using + sign. In this way u can add multiple questions.

The screenshot shows the form with two questions. The first question is "Which of the following are the features of cloud computing?" with the same options as before. The second question is "Which of the following is a type of cloud computing service?" with a "Multiple choice" dropdown. The second question has five options: "Service-as-a-Software (SaaS)", "Software-and-a-Server (SaaS)", "Software-as-a-Service (SaaS)", "Software-as-a-Server (SaaS)", and "Add option or Add 'Other'". The "Software-as-a-Service (SaaS)" option is selected with a green checkmark. The "Required" toggle switch for the second question is also turned ON.

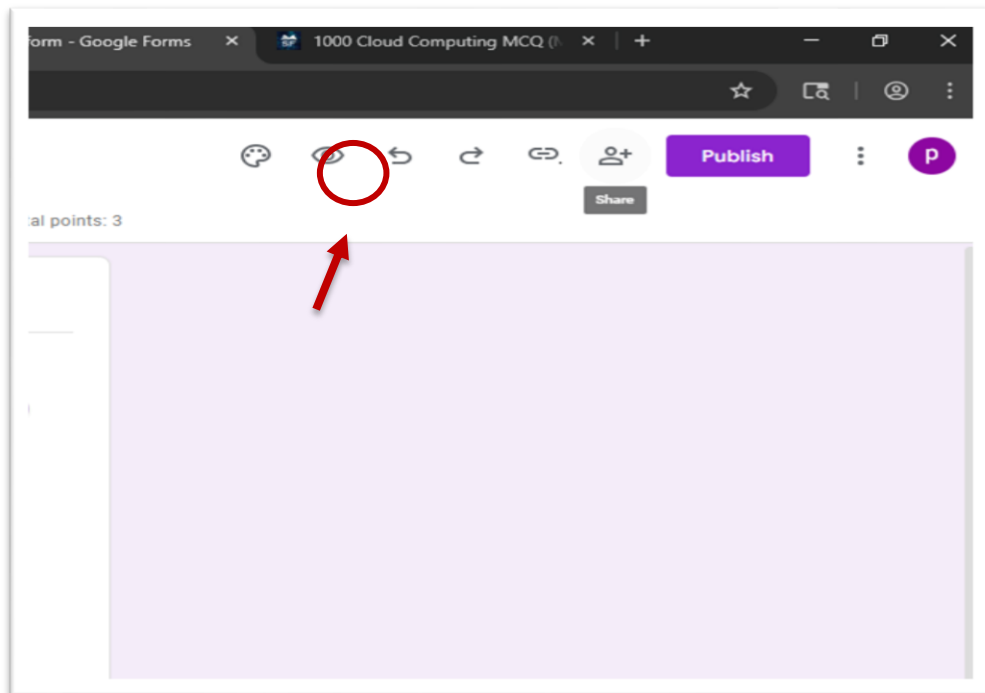
10.After completion of question go to **setting**, select **RELEASE MARKS** either **immediately** or **later**.



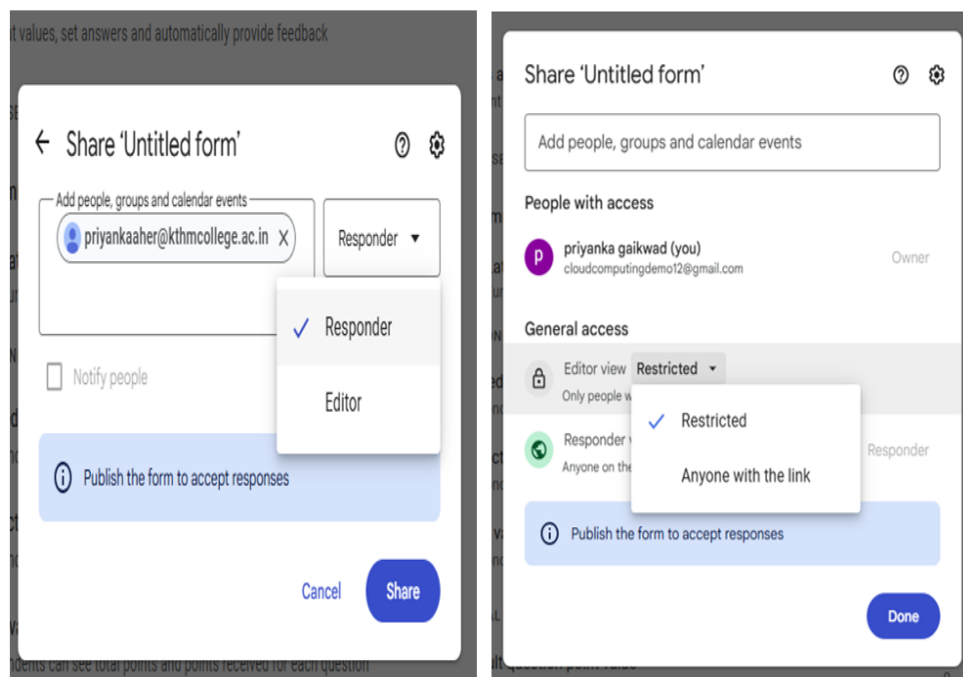
11. To generate the form link, click on **Publish** and **copy** the link (You can get short link also using **shorten URL**).



12. Share the link using **share** option



13. After click on **share**, you can add people and give **general** access to **editor** and **responder** view either **Restricted** or **Anyone with the link**.



14. Paste the link in **message** box Share the link using **Send** button.

← Share 'Untitled form' ? ⚙

Add people, groups and calendar events

priyankaaher@kthmcollege.ac.in ✕

Responder ▾

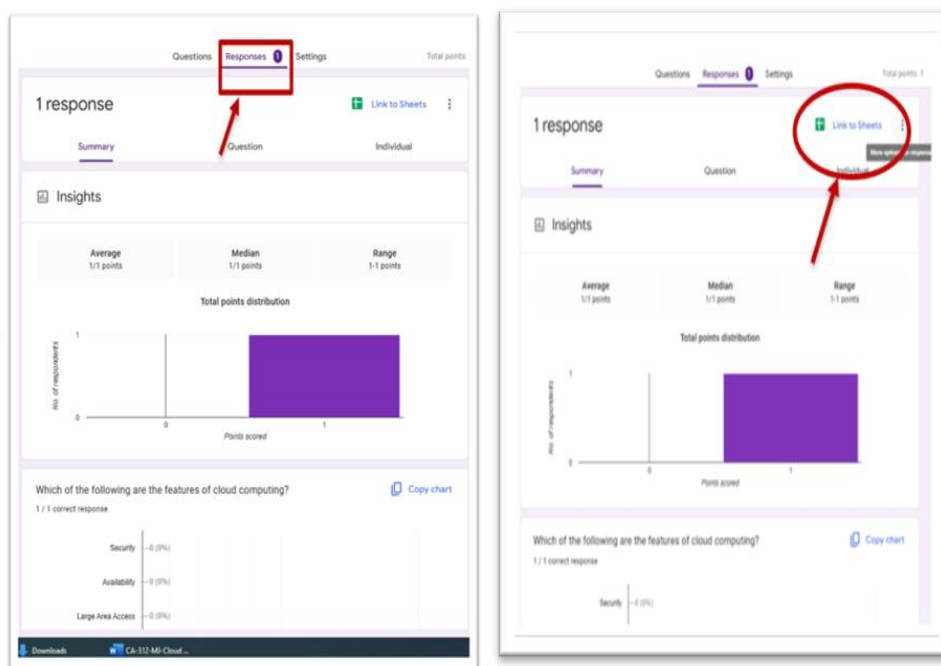
☒ Notify people
Form will be included in the email notification

Message

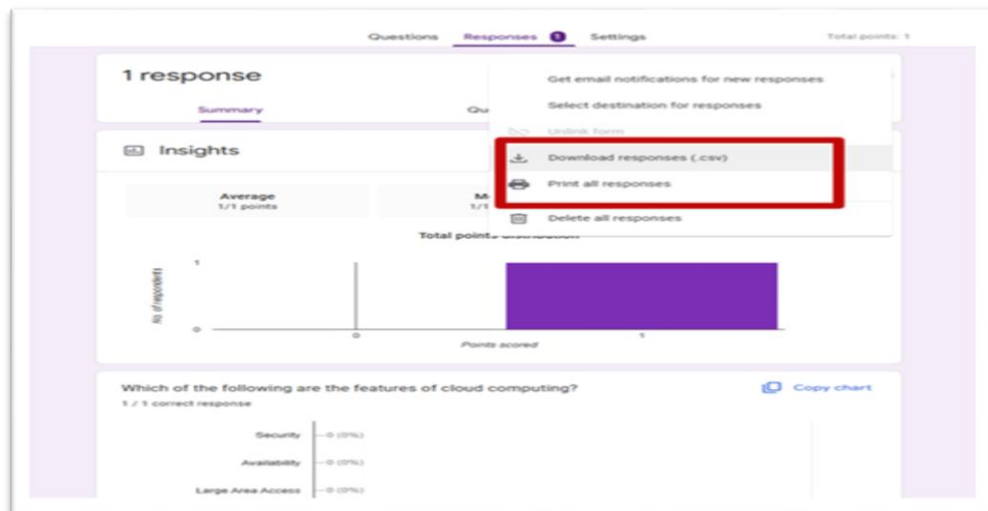
<https://forms.gle/1MxcJuZXWQfocftj8>

🔗 Cancel Send

15. You see the response using Responses tab and download response using Link to Sheets.



16. After click on three dots u can download or print responses and analyze result using responses.



Lab Assignment

Set A

1. working of Google Drive in creating spreadsheets and notes.
2. How is Storage as a Service (STaaS) implemented? Explain with a practical example.
3. Practical implementation of file sharing and storage as a service.

Set B

1. Working of google drive to create exam form and generate result.
2. Create google form for generate survey about beauty products.
3. Working with google drive to make spread sheet (for 10 students record).
4. Working with google drive to make spread sheet (purchase order for grocery items).

Set C

1. Create google form for accepts details of student and create test page and generate result.
2. Create a google form for generating certificates for workshops.

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Needs Improvement []

4: Complete []

5: Well Done []

Signature of Instructor:

Date

Assignment No.2

Title: Understanding Virtualization using Virtual box

Prerequisite:

- A computer running Windows Server or Windows client with Hyper-V enabled.
- Sufficient physical memory on the host to allocate to the virtual machine
- An operating system installation media (.iso file) or an existing virtual hard disk (.vhd or .vhdx) if you want to install an operating system.
- Membership in the local Administrators group or the Hyper-V Administrators group.

Guidelines for Teachers/Instructors:

- Prepare workspace before assignment
- Clearly define assignment objectives
- Set permissions and access control carefully
- Shares file properly.
- Use either Hyper-V Manager or Windows PowerShell.
- Select a virtual switch configured if you want network connectivity for the virtual machine.

Instructions for Students: -

- Students must have a valid Google account.
- Carefully read instructions provided by the teacher.
- Understand the objective of assignment clearly.
- Solve Set A, B and C assigned by instructor in allocated slots only.

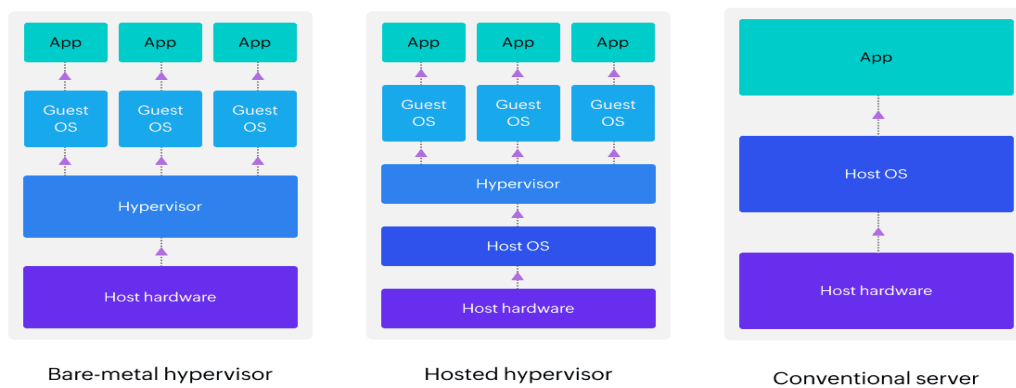
1. Introduction

○ **What is Virtualization?**

Virtualization in cloud computing is a technology that allows you to create virtual (software-based) versions of physical resources like servers, storage devices, networks, or even entire operating systems.

Instead of using one physical machine for one task, virtualization lets a single machine run multiple independent environments at the same time. Virtualization is powered by a software layer called a hypervisor. The hypervisor sits between hardware and virtual machines. It allocates resources like CPU, memory, and storage. Each virtual machine (VM) runs its own OS and applications independently.

Virtualization architecture



○ Types of Virtualization

➤ Serve Virtualization

Splits one physical server into multiple virtual servers

➤ Storage Virtualization

Combines multiple storage devices into one virtual storage system

➤ Network Virtualization

Creates virtual networks separate from physical hardware

➤ Desktop Virtualization

Runs desktop environments on a central server instead of local machines

○ Characteristics of virtualization

- **Abstraction** – Hides physical hardware using tools like Oracle VM Virtual Box
- **Isolation** – Each VM works independently without affecting others
- **Encapsulation** – VM is stored as a single file for easy management
- **Resource Sharing** – Multiple VMs share the same hardware resources
- **Scalability** – Resources and VMs can be easily increased or decreased
- **Portability** – VMs can be moved across systems (e.g., via **VMware Workstation**)
- **Flexibility** – Run different OS like **Ubuntu** and **Windows 11** together
- **Snapshots** – Save and restore VM states anytime
- **Hardware Independence** – Not dependent on specific physical hardware
- **Security** – Provides a safe and isolated environment

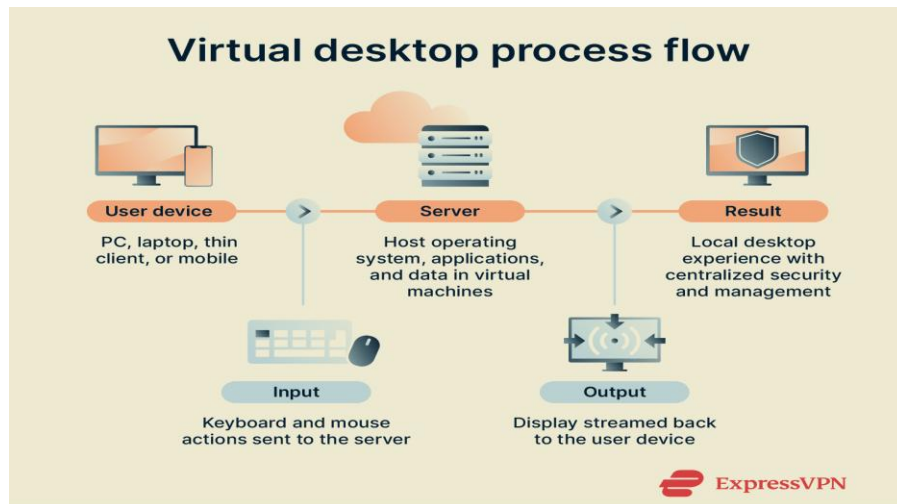
Create Desktop Virtualization:-

Desktop virtualization decouples a user's operating system, applications, and data from their physical computer, running them instead as virtual machines (VMs) on a centralized server or in the cloud.

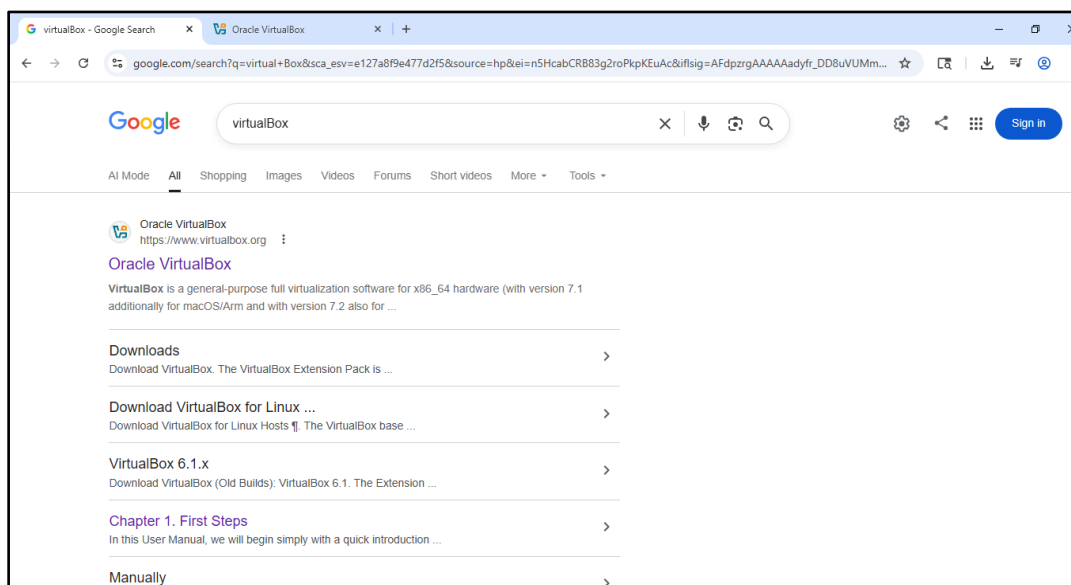
Desktop as a Service (DaaS) is a form of desktop virtualization that is entirely hosted in the cloud by a third-party provider.

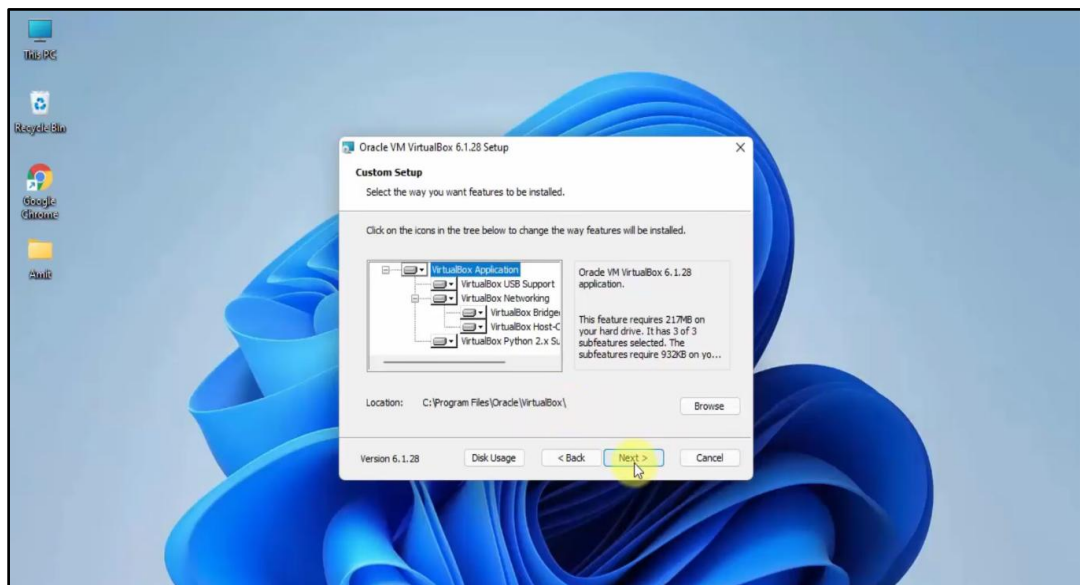
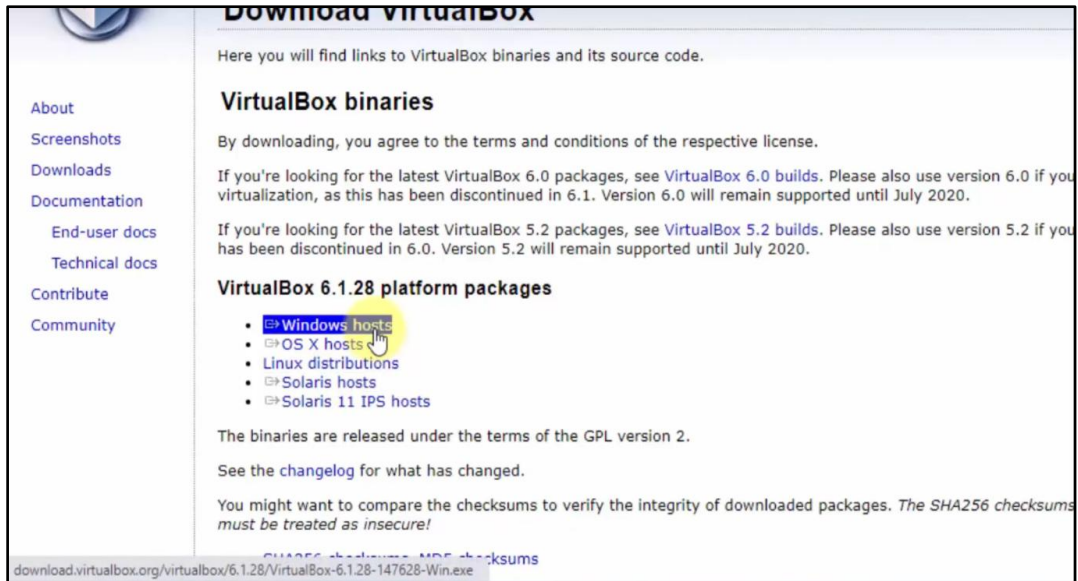
Session-based virtualization, also known as RDS (Remote Desktop Services) or session-based computing, is based on the principle of resource sharing.

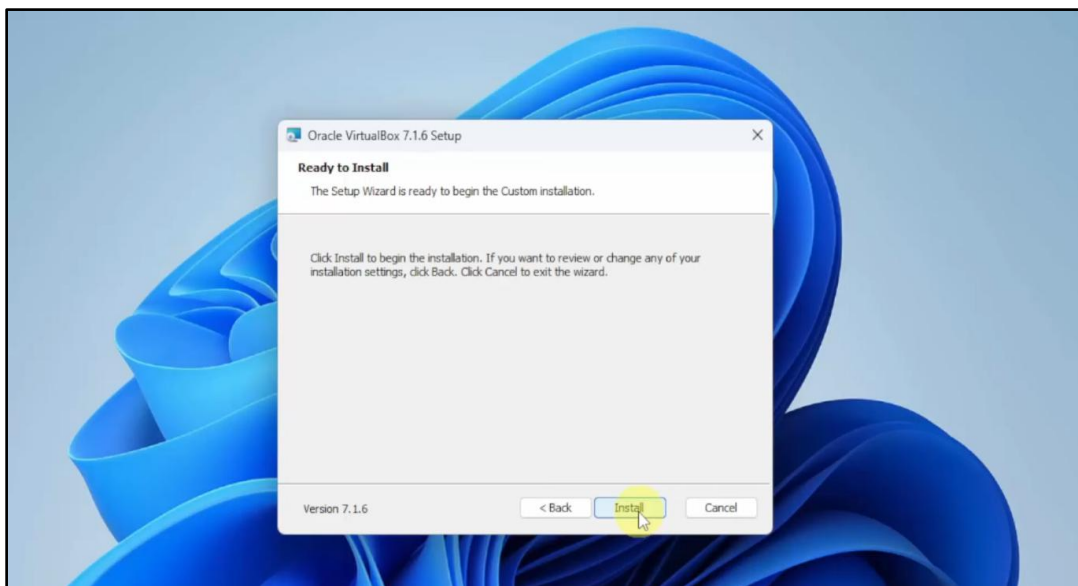
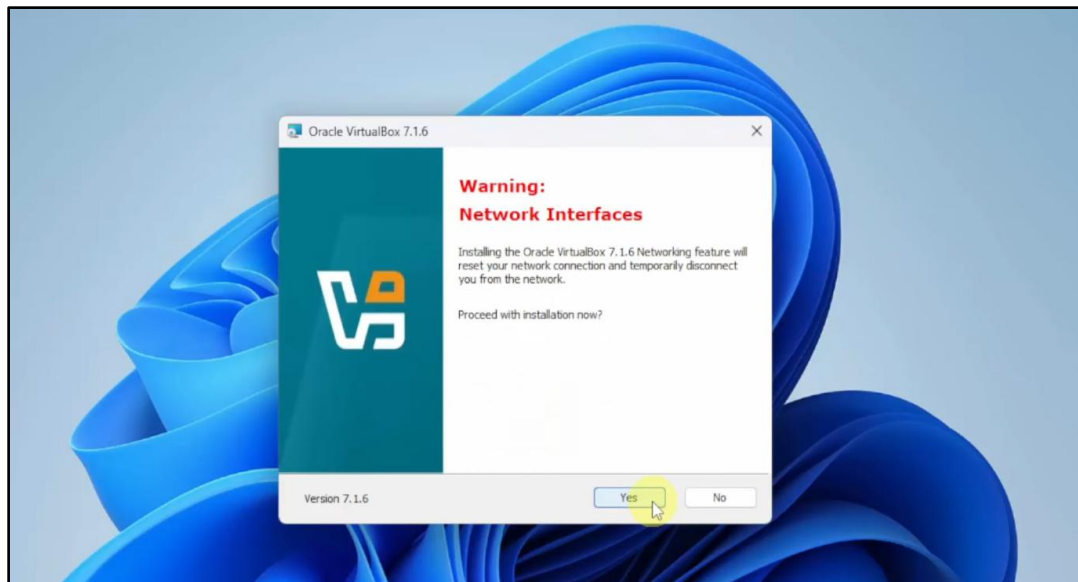
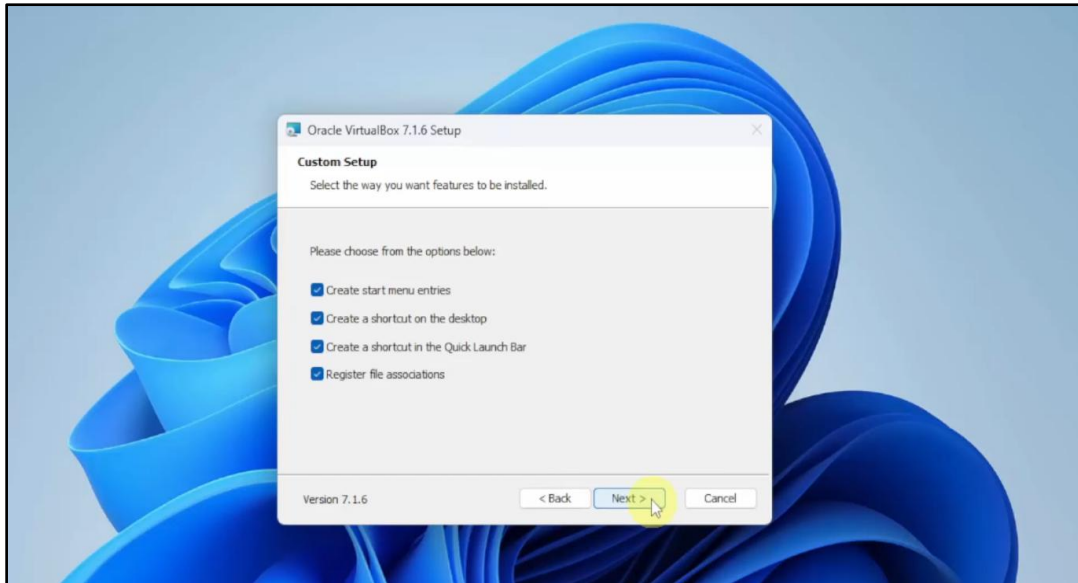
In the case of local virtualization, the virtual environment is hosted directly on the user's computer using a virtual machine.

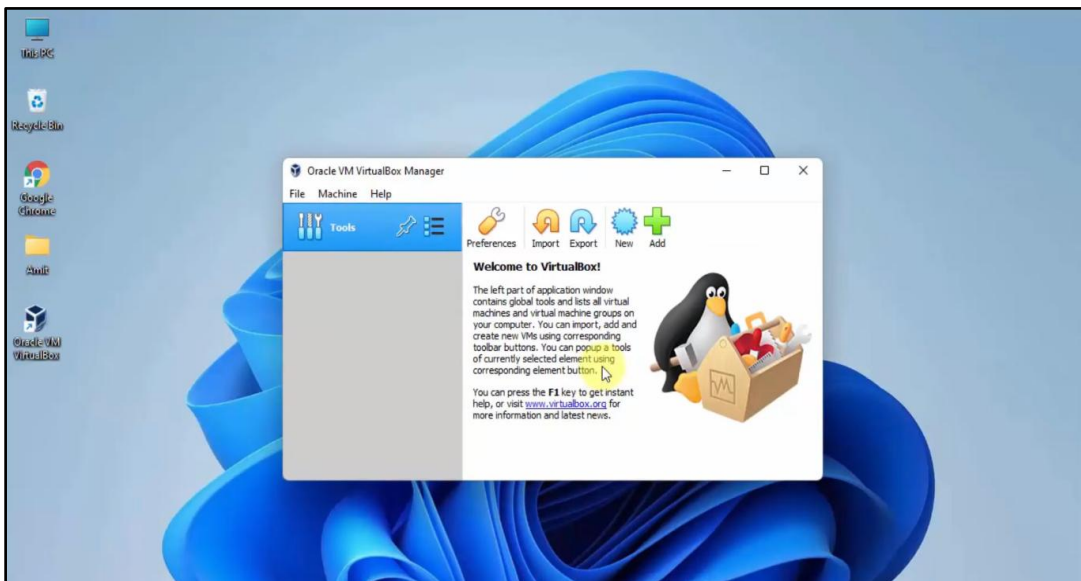
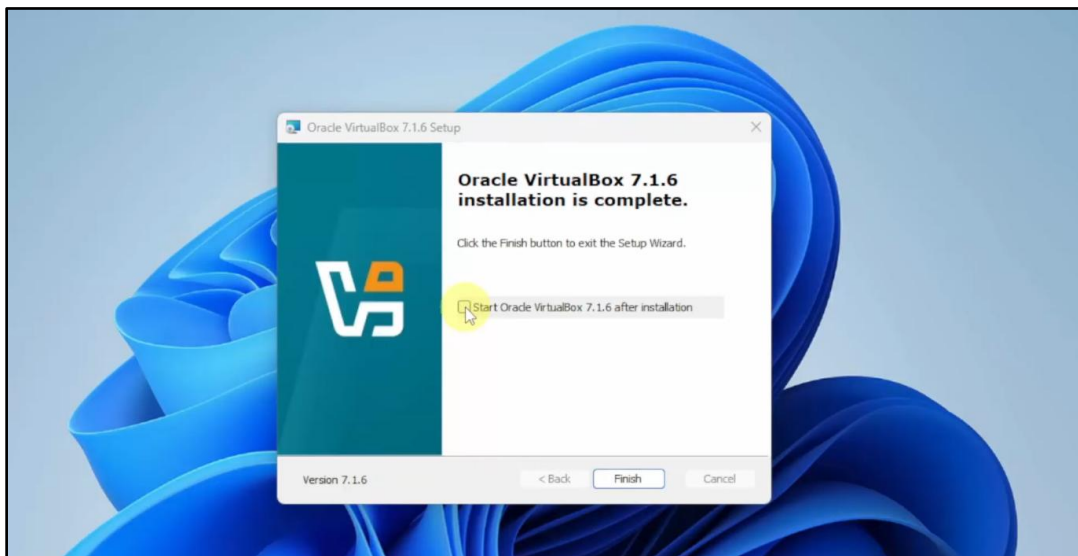
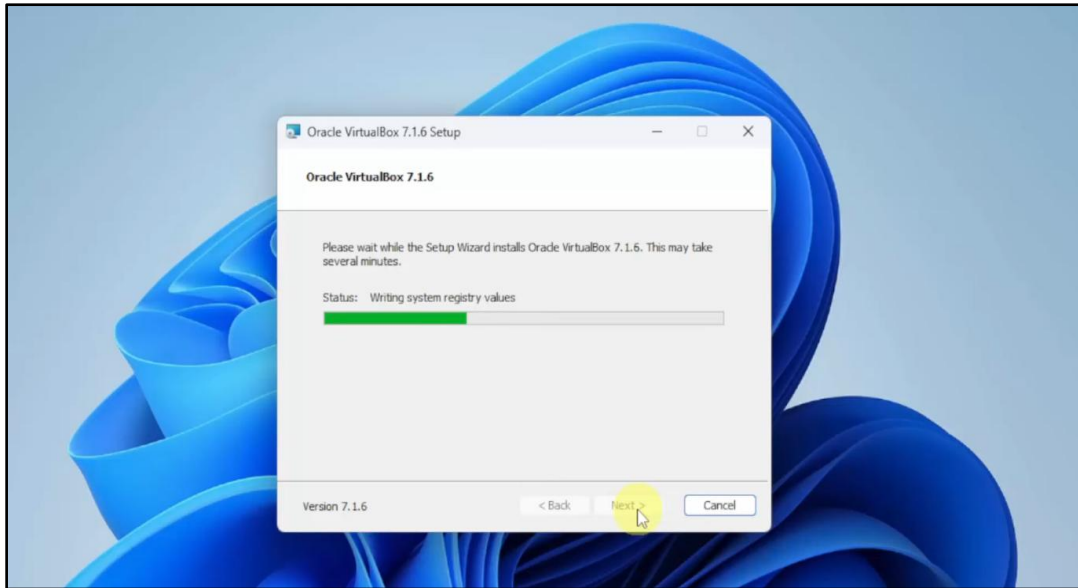


How to install Virtual Machine on windows OS:-

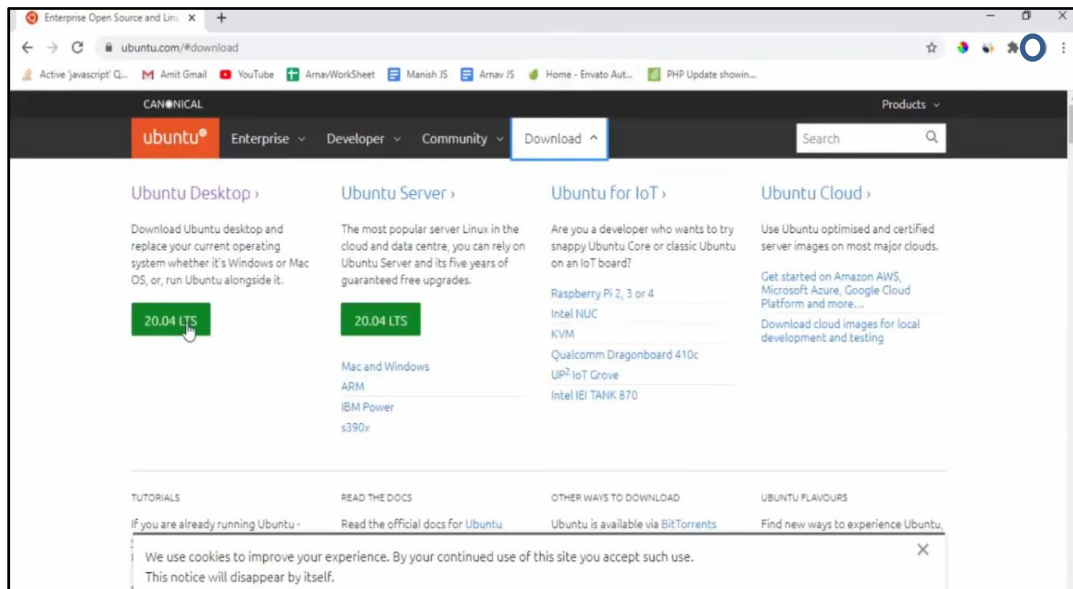
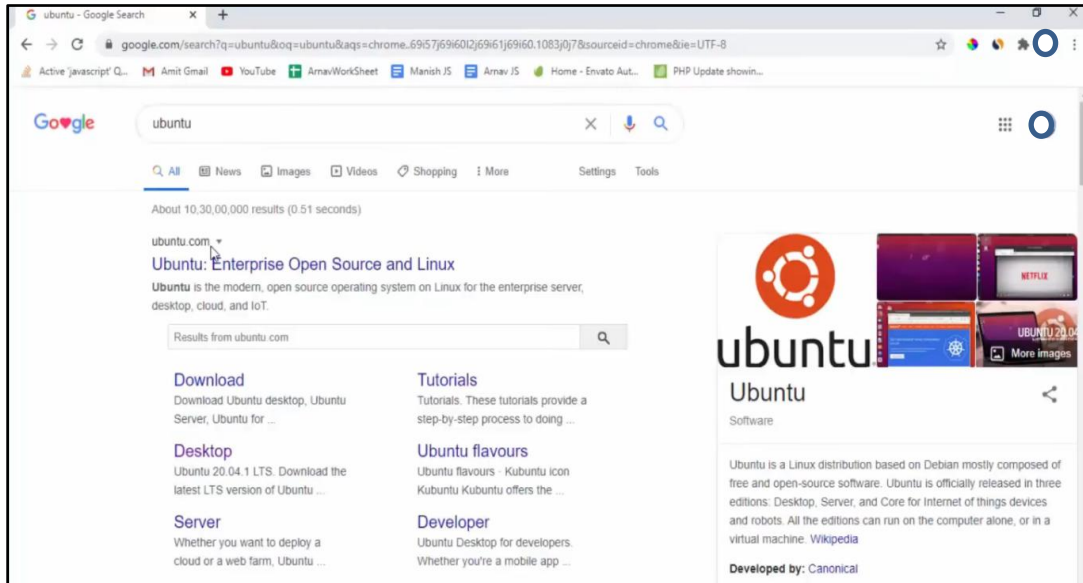




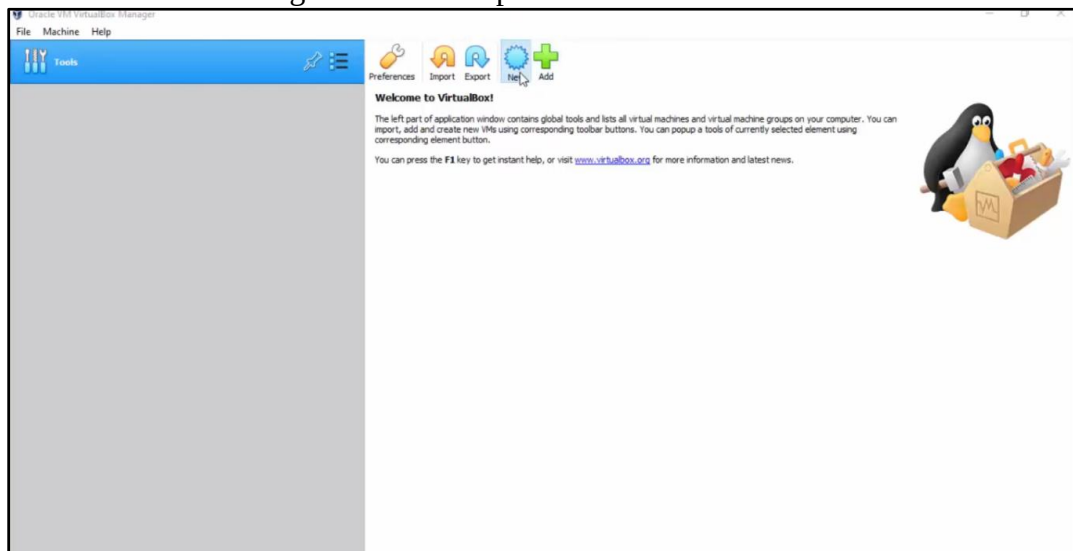




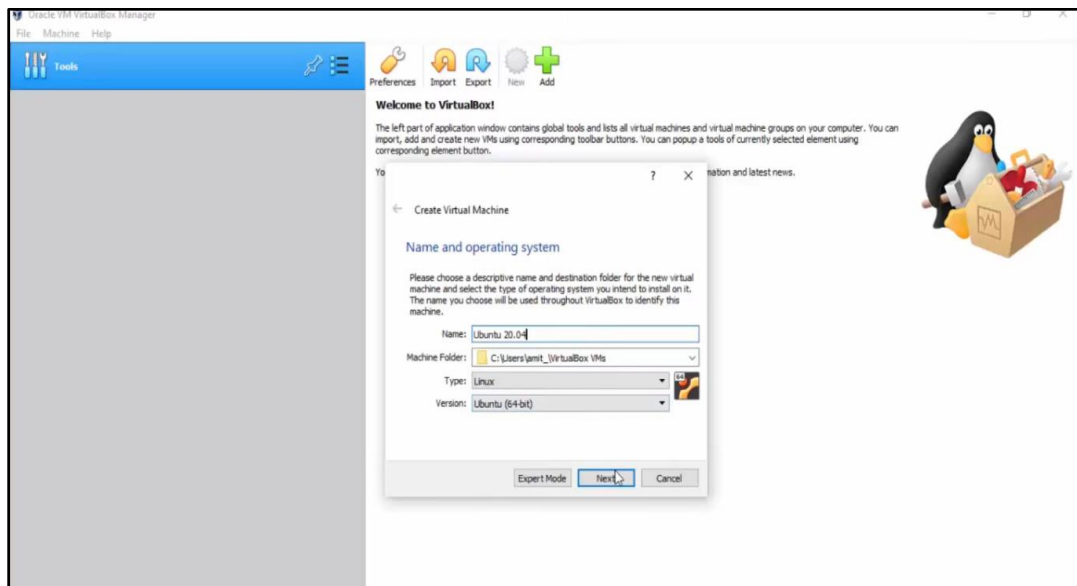
Install Ubuntu OS in Virtual Box



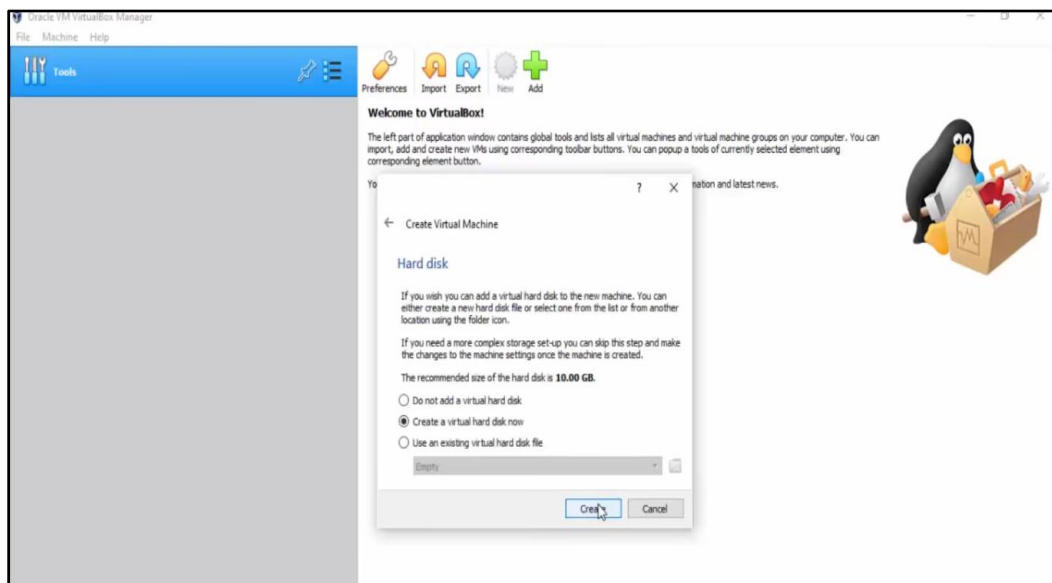
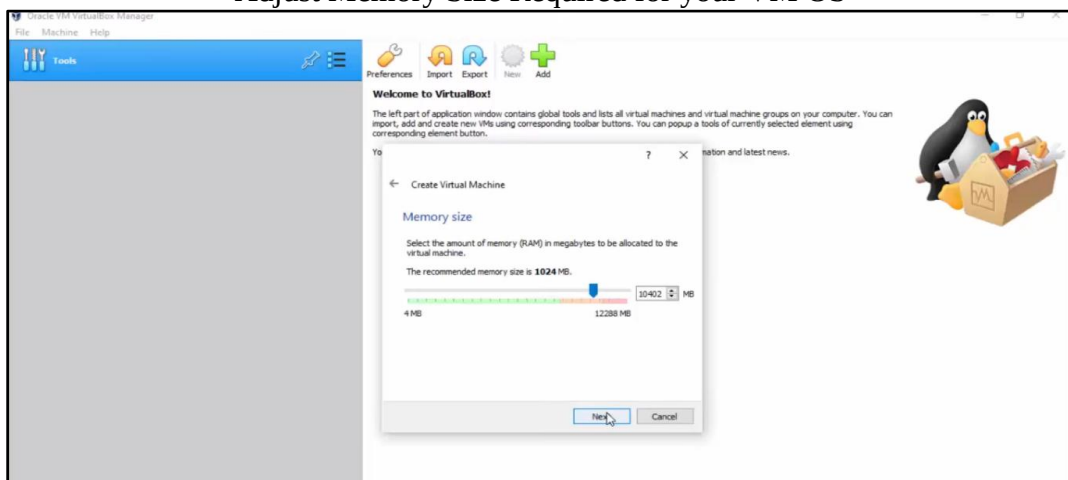
After Downloading Ubuntu OS. Open Oracle VM then click on New Button

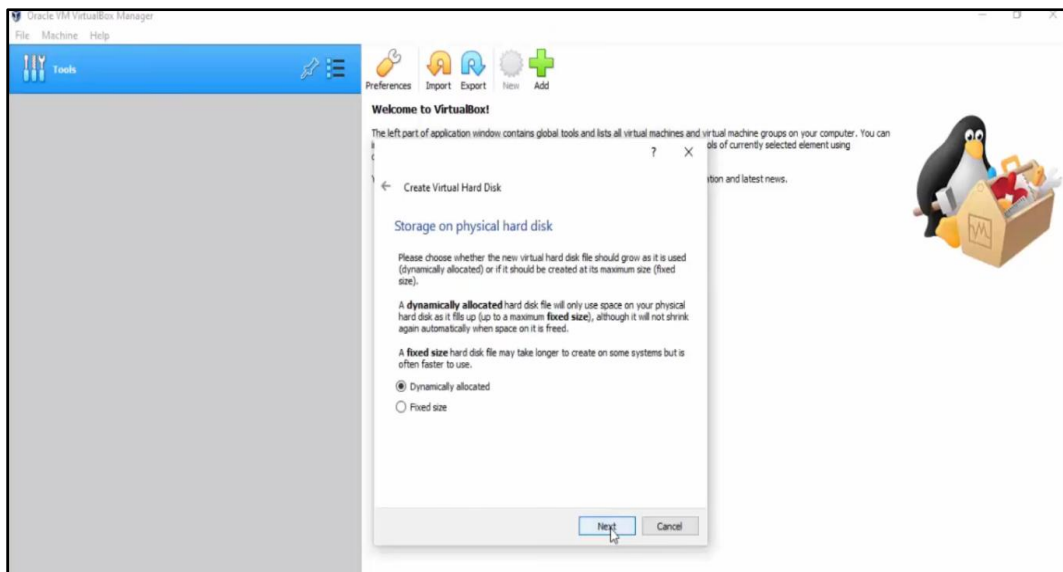


Open new Window “Create Virtual Machine” then type into first textfield OS name and version

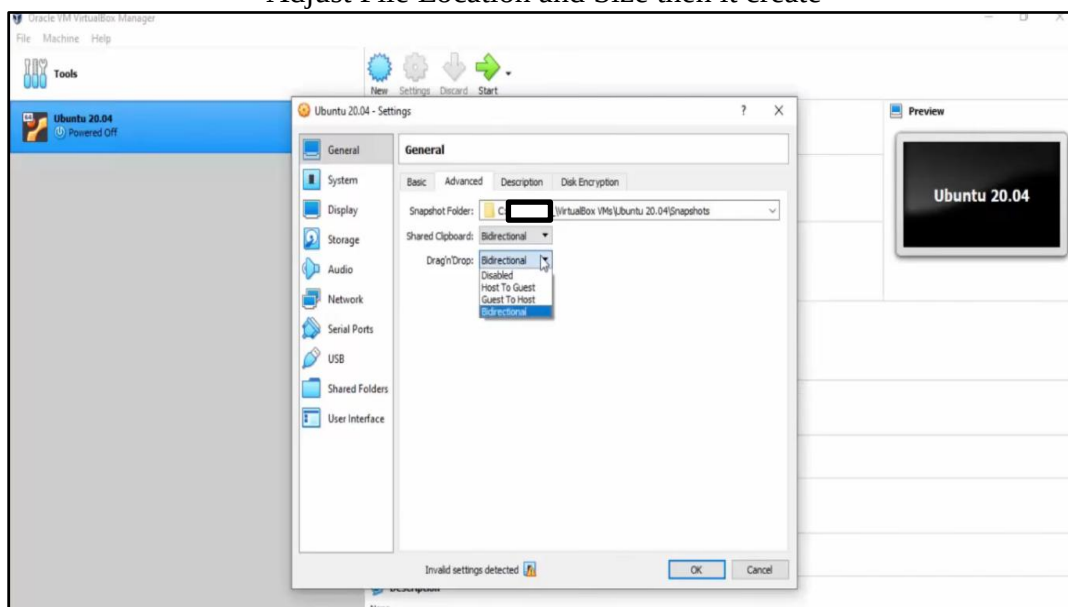


Adjust Memory Size Required for your VM OS

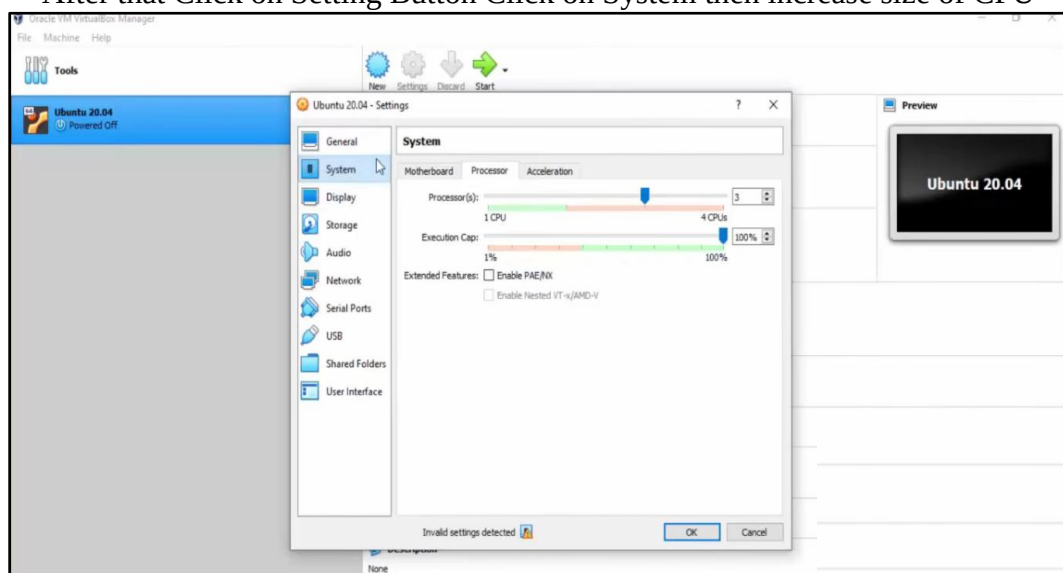


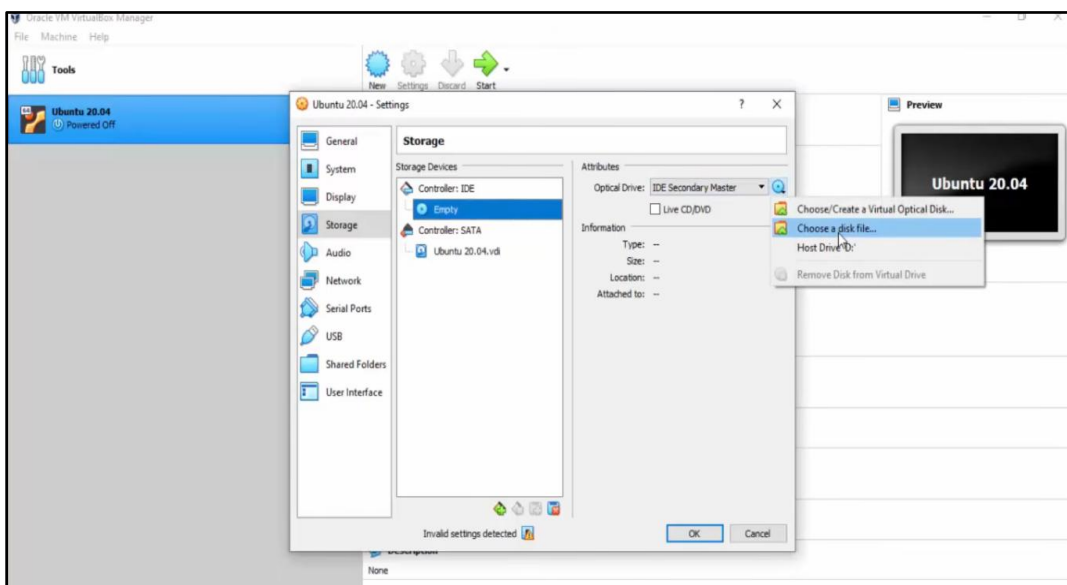
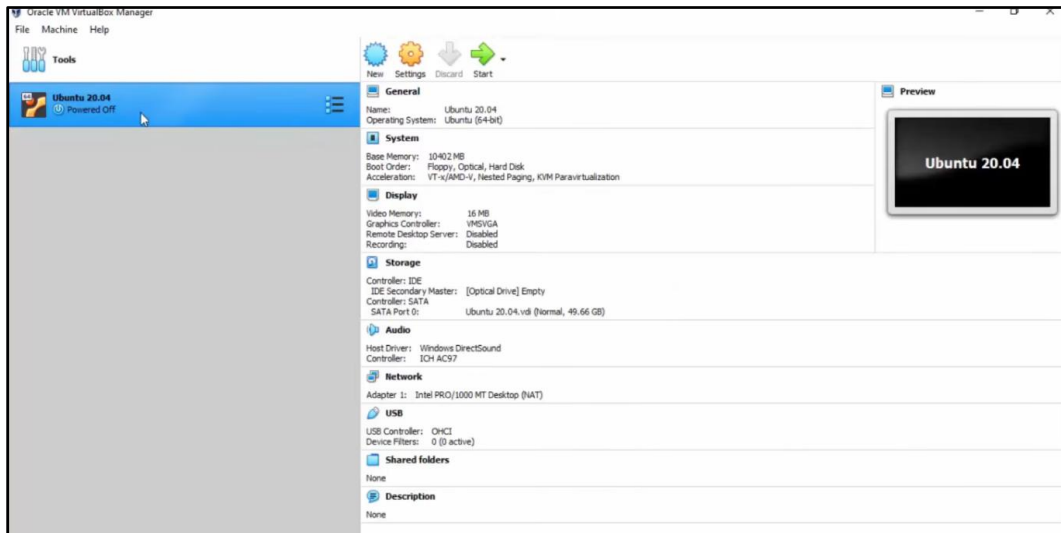


Adjust File Location and Size then it create

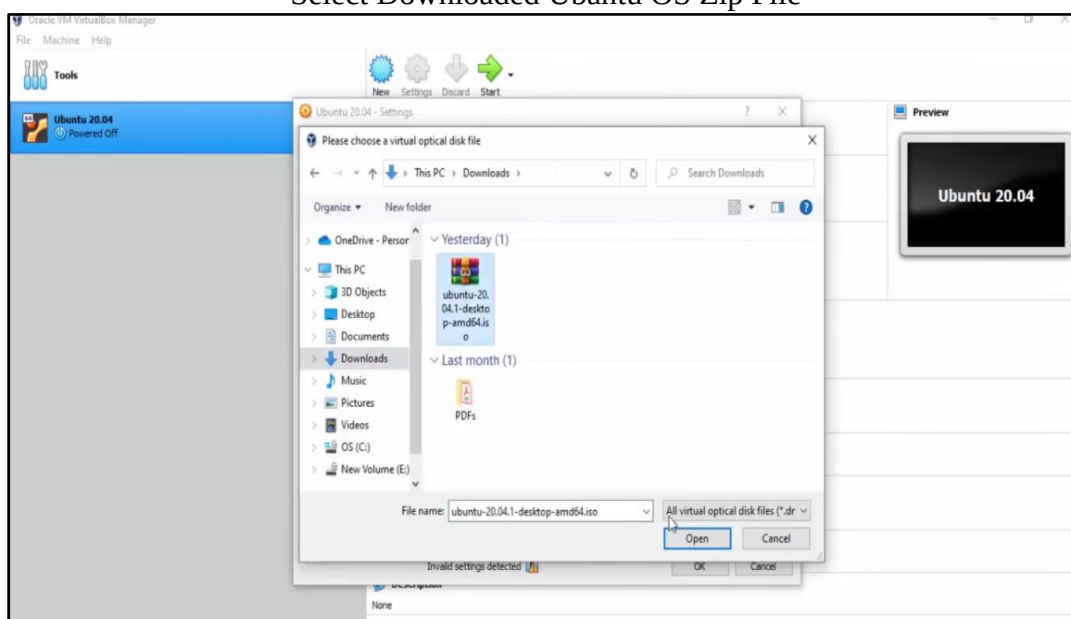


After that Click on Setting Button Click on System then increase size of CPU

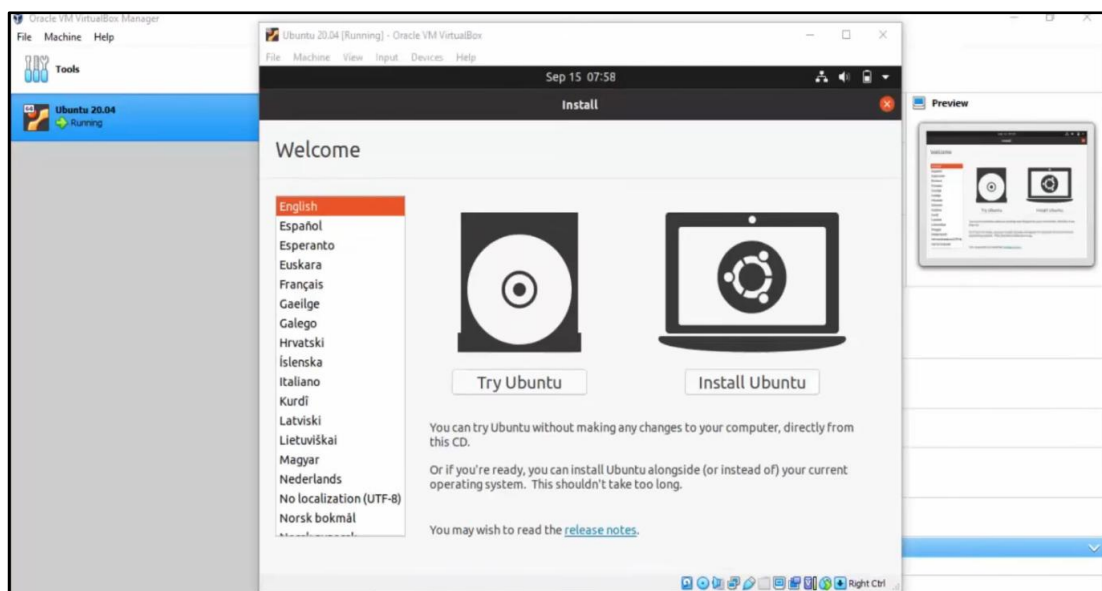
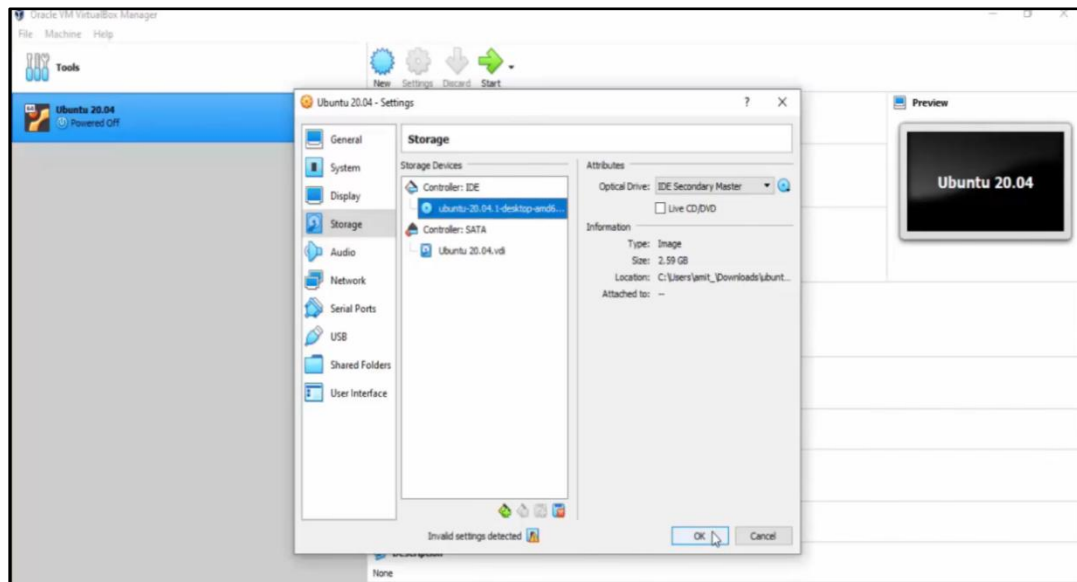
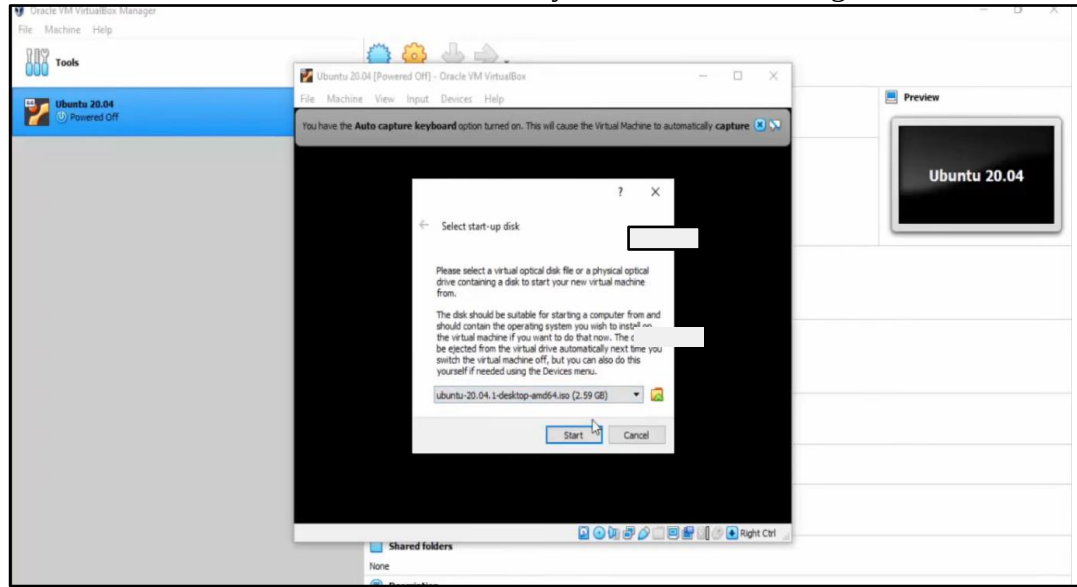


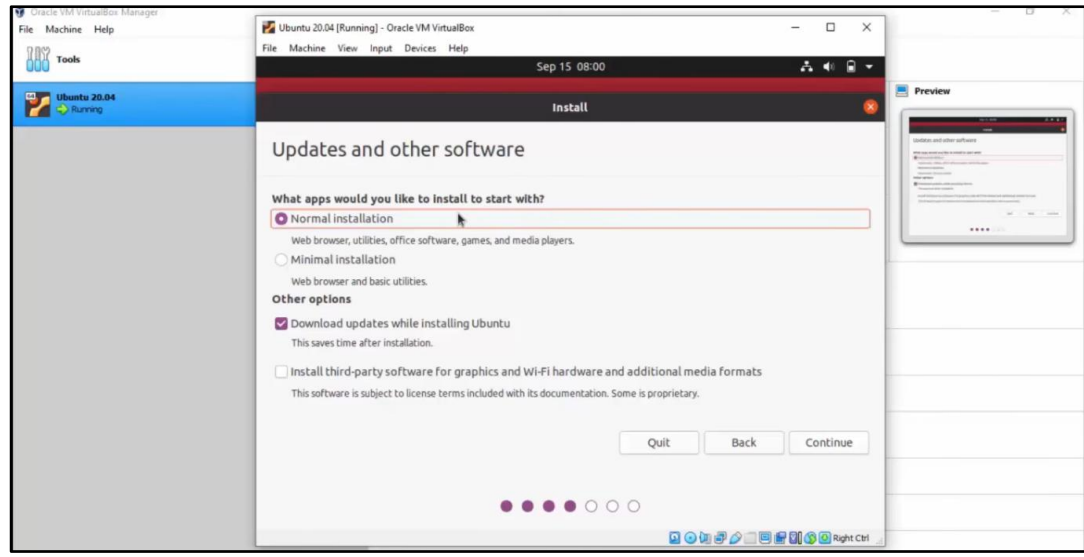


Select Downloaded Ubuntu OS Zip File

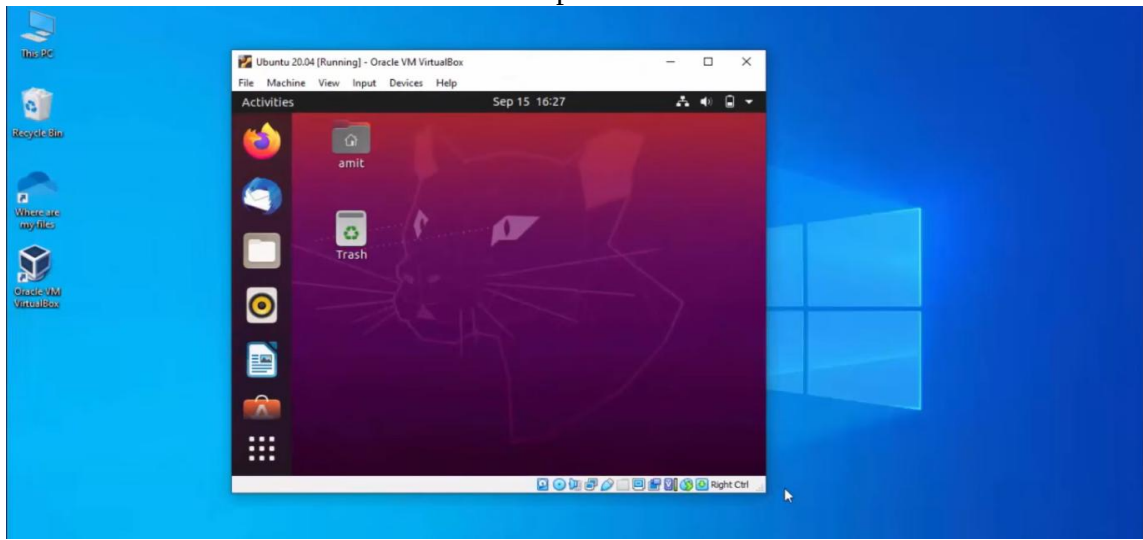


Click on Start Button on VM window after that you will Ubuntu OS again Click on start Button





Install As per instructio



Lab Assignment

Set A

1. Create Virtual Machine.
2. Create Virtual Machine and Perform Basic Shell Commands.

Set B

1. Create Virtual Machine and Install Ubuntu OS.
2. Create Virtual Machine and Install Oracle Database and Execute Basic Query Processing.

Set C

1. Create Virtual Machine and Linux OS.

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Needs Improvement []

4: Complete []

5: Well Done []

Signature of Instructor :

Date:

Assignment No.3

Title: Introduction to Google Cloud Console & Cloud Shell Editor

Prerequisite:-

- Basic knowledge of Python Programming
- A Google Account (Gmail) to access Google Cloud
- Basic understanding of cloud computing concepts

Guidelines for Teachers/Instructors:-

- Explain what Google Cloud Platform (GCP) is and its key services
- Demonstrate how to log in to the Google Cloud Console
- Show students how to activate Cloud Shell and open the Cloud Shell Editor
- Explain the concept of a project in GCP and how billing works (free tier)

Instructions for Students:

- Use your personal or college Google account to access Google Cloud Console
- Do NOT share your account credentials with anyone
- Every student must perform the steps individually on their own account
- Solve Set A, B and C as assigned by the instructor in allocated slots only

Introduction:

Google Cloud Platform (GCP) is a suite of cloud computing services offered by Google. It provides a range of hosted services for computing, storage, networking, machine learning, and application development that run on the same infrastructure that Google uses internally for its end-user products such as Google Search, Gmail, and YouTube.

What is Google Cloud Console?

The Google Cloud Console is a web-based graphical user interface (GUI) that allows you to manage your Google Cloud resources. You can create and manage virtual machines, set up databases, deploy applications, monitor services, and configure billing — all from your browser without installing any software.

What is Cloud Shell?

Google Cloud Shell is a free, browser-based command-line environment that comes pre-installed with all the tools you need to manage your GCP resources — including the Google Cloud CLI (**gcloud**), Python, Node.js, and many other languages. It provides 5 GB of persistent home directory storage and runs on a Debian-based Linux virtual machine.

What is the Cloud Shell Editor?

The Cloud Shell Editor is a full-featured, browser-based code editor built on top of Code OSS

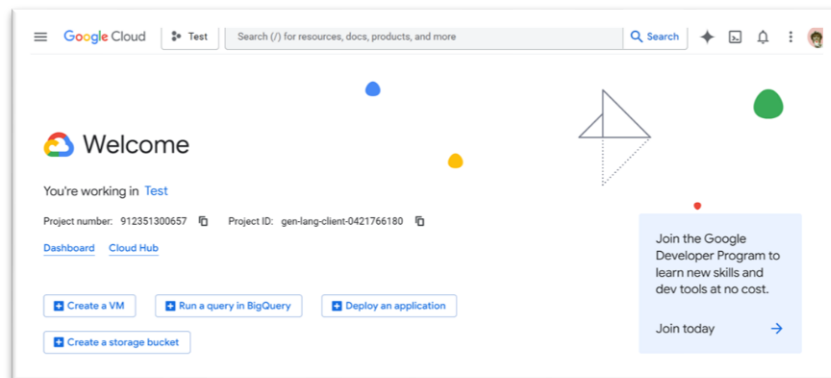
(the open-source base of Visual Studio Code). It provides syntax highlighting, IntelliSense, file explorer, integrated terminal, and extensions — making it feel just like working in VS Code, but entirely in your browser with no installation needed.

Accessing Google Cloud Console

Step 1 – Open the Console

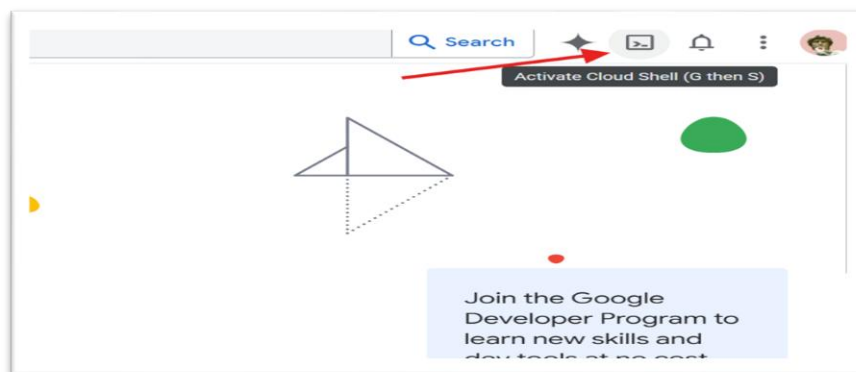
Open your browser and navigate to: **<https://console.cloud.google.com>**

Sign in using your Google account. You will land on the Cloud Console Dashboard.

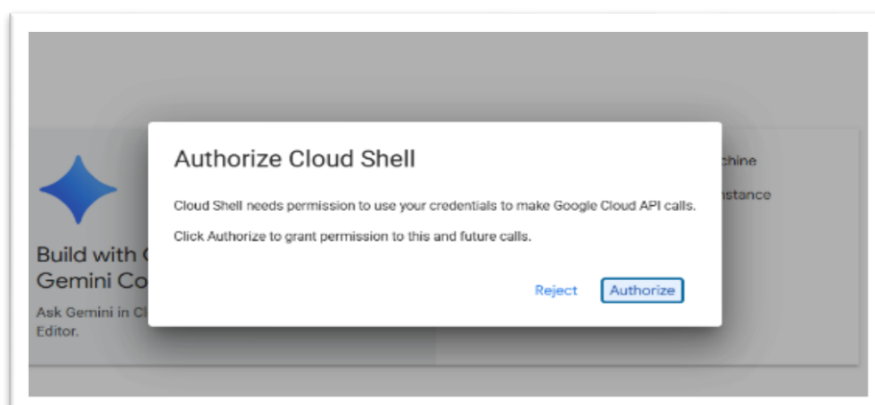


Step 2 – Activate Cloud Shell

In the top-right corner of the Console, click the Activate Cloud Shell icon (it looks like a terminal prompt: >_). A terminal panel will open at the bottom of your browser window.



Note: The first time you open Cloud Shell, it may take about 30–60 seconds to provision your virtual machine. This is normal.



Step 3 – Verify Python is Available

```
python3 --version
```

In the Cloud Shell terminal, type the following command and press Enter:

You should see an output like **Python 3.x.x**, confirming Python is pre-installed.

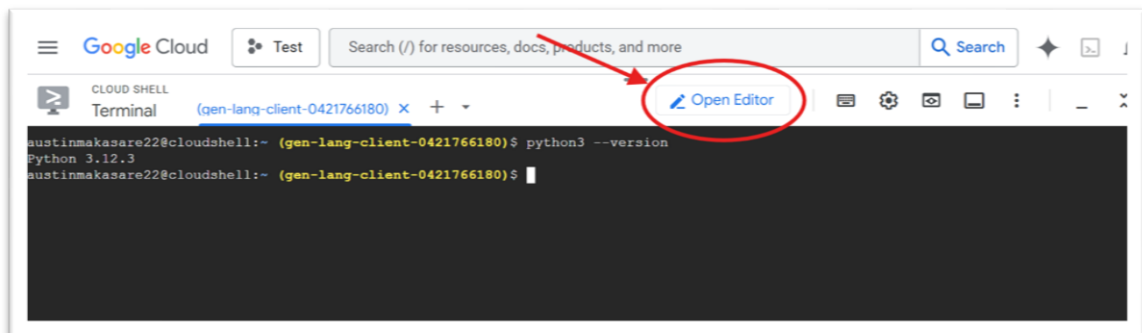


Opening the Cloud Shell Editor

Step 1 – Launch the Editor

In the Cloud Shell toolbar (top of the terminal panel), click the **Open Editor** button (pencil icon).

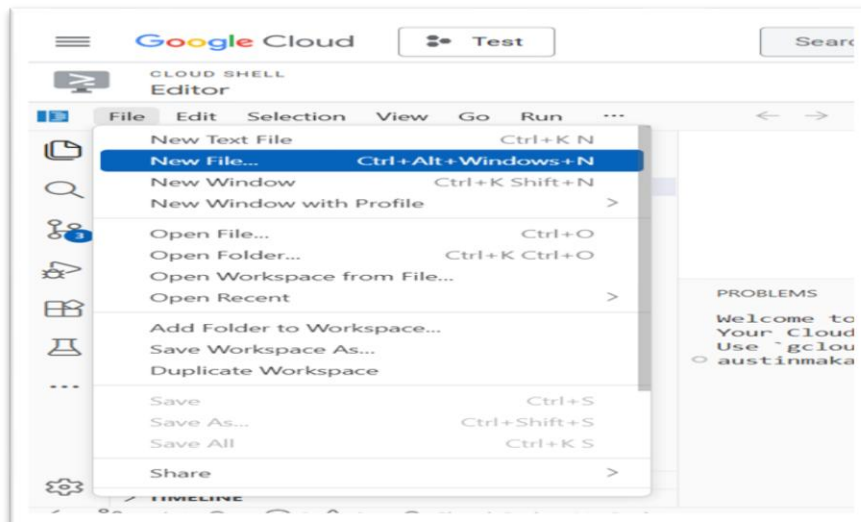
The editor will open in a new browser tab and will look very similar to Visual Studio Code.



Step 2 – Create a New Python File

In the Editor, go to File-New File. Name the file hello.py and save it in your home directory.

You can also use the integrated terminal inside the editor by clicking Terminal-New Terminal.



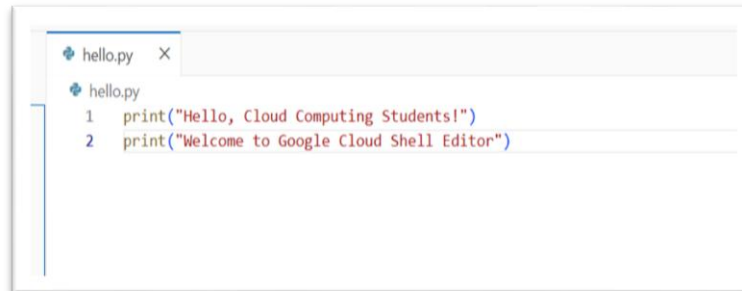
Step 3 – Switch Between Editor and Terminal

You can toggle between the Cloud Shell Editor and the terminal at any time using the **Open Terminal** and **Open Editor** buttons. Both share the same file system so files created in the editor are immediately accessible in the terminal.

Writing and Running Python Programs

Your First Python Program in Cloud Shell

Type the following program in the Cloud Shell Editor and save it as **hello.py**:



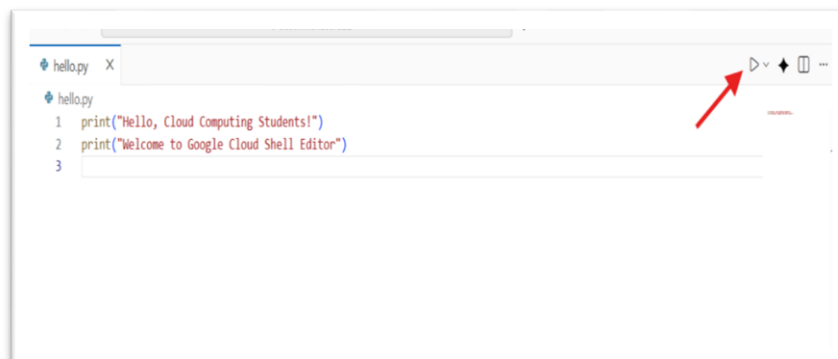
```
hello.py x
hello.py
1 print("Hello, Cloud Computing Students!")
2 print("Welcome to Google Cloud Shell Editor")
```

To run the program,

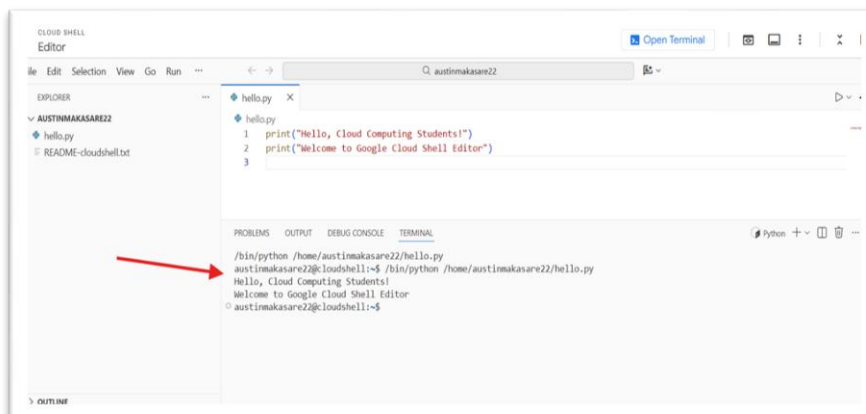
1. open the terminal and type:

```
python3 hello.py
```

2. Direct run by clicking the run button:



You will see the output printed directly in the terminal window.



Run the program using:

```
python3 add.py
```

Accepting Input from the User

The **input()** function in Python reads a value entered by the user from the terminal. Consider the following program that accepts two numbers and adds them. Save it as **add.py**:

```
# add.py – Accept two numbers and display their sum
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = a + b
```

Creating and Managing Files from the Terminal

Cloud Shell provides a standard Linux terminal. You can use common Linux commands to manage files alongside your Python scripts:

ls — List all files in the current directory

mkdir myfolder — Create a new folder named myfolder

cd myfolder — Navigate into the folder

touch myfile.py — Create a new empty Python file

cat myfile.py — Display the contents of a file

python3 myfile.py — Execute a Python script

Practice Programs

1. Write a Python program to print your name and college name using `print()` statements.
2. Write a Python program to find the area and perimeter of a rectangle.
3. Write a Python program to check if a number entered by the user is even or odd.
4. Write a Python program to find the factorial of a number.
5. Write a Python program to reverse a string entered by the user.
6. Write a Python program to check whether a number is a palindrome or not.

Lab Assignment

Set A

1. Write a Python program in Cloud Shell Editor to print the Fibonacci series up to n terms (accept n from user).
2. Write a Python program to generate and print the multiplication table (1 to 10) using nested loops.
3. Write a Python program using `if-elif-else` that accepts two numbers and an arithmetic operator (+, -, *, /) from the user and displays the result.
4. Write a Python program to check whether a number entered by the user is a palindrome.

Set B

1. Write a Python program in Cloud Shell Editor to add two complex numbers entered by the user and display the result.
2. Write a Python program to calculate the following series and display the sum:
 $(1) + (1+2) + (1+2+3) + \dots + (1+2+3+\dots+n)$

3. Write a Python program to convert a decimal number entered by the user into its hexadecimal equivalent without using built-in functions.

Set C

Write a Python program in Cloud Shell Editor that displays the following star pattern:

```
*
* *
* * *
* *
*
```

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Needs Improvement []

4: Complete []

5: Well Done []

Signature of Instructor :

Date

Assignment No: 4

Title: Create AWS and Azure Account

Prerequisite:

- Basic understanding of Computer Networks and Operating Systems.
- Sufficient physical memory on the Hardware.
- Accounts required 1) AWS Free Tier account (via Amazon Web Services)
2) Azure Free account (via Microsoft).
- Students having valid Email and Bank Debit Card and PAN card details.
- Students having knowledge Navigate AWS and Azure dashboards.

Guidelines for Teachers/Instructors:

- Ensure students can access Amazon Web Services and Microsoft Azure accounts.
- Introduce cloud basics, demonstrate AWS EC2 and Azure VM creation.
- Teacher must be monitor for errors and unnecessary costs.
- Ensure students use free-tier services, stop resources after use, and avoid high-cost configurations.
- Clearly define assignment objectives
- Set permissions and access control carefully

Instructions for Students: -

- Students must have a valid Gmail account, PAN card and Bank debit card details.
- Carefully read instructions provided by the teacher.
- Understand the objective of assignment clearly.
- Solve Set A, B and C assigned by instructor in allocated slots only.

2. Introduction:

○ What is AWS and Azure?

AWS is a cloud computing platform provided by Amazon that offers a wide range of on-demand services like computing power, storage, databases, and networking over the internet.

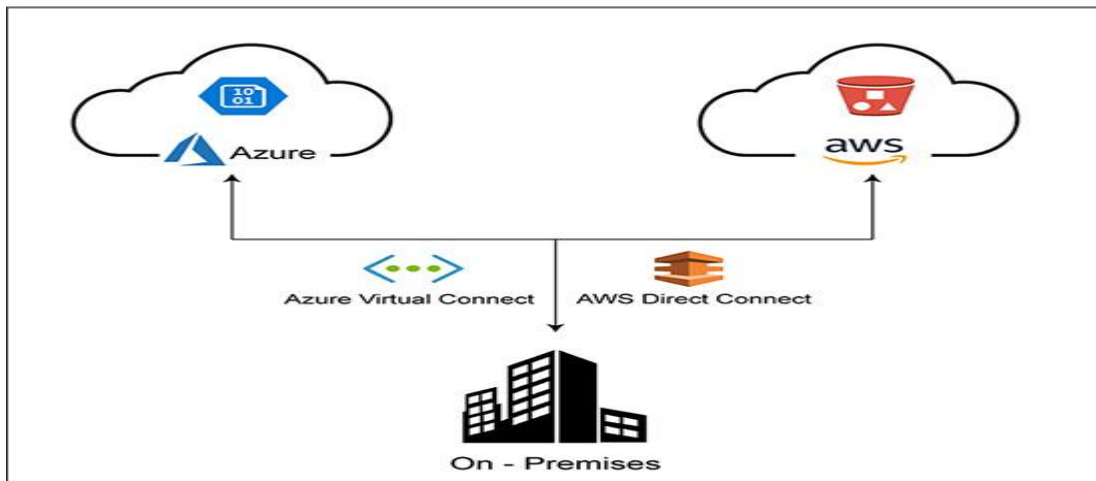
Azure allows users to build, deploy, and manage applications in the cloud without needing physical servers or infrastructure. EC2 (virtual servers), AWS Lambda (serverless computing), Azure Virtual Machines, Virtual Network (VNet).

○ Uses of AWS and Azure account:

- Host websites and applications on the internet
- Store and back up data securely in the cloud
- Create virtual machines (cloud-based computers)
- Develop and deploy software applications
- Run big data analytics and processing

- Build AI and machine learning models
- Enable remote access and collaboration
- Disaster recovery and system backup

Multi-Cloud Hybrid Architecture



Types of Amazon Web Services (AWS) Accounts-

a) Free Tier Account

Provides limited free usage for 12 months (and some always-free services)

Suitable for students, beginners, and practice labs

b) Pay-As-You-Go (Standard Account)

Users pay only for the services they use

No fixed cost; billing depends on usage

c) Enterprise Account

Designed for large organizations

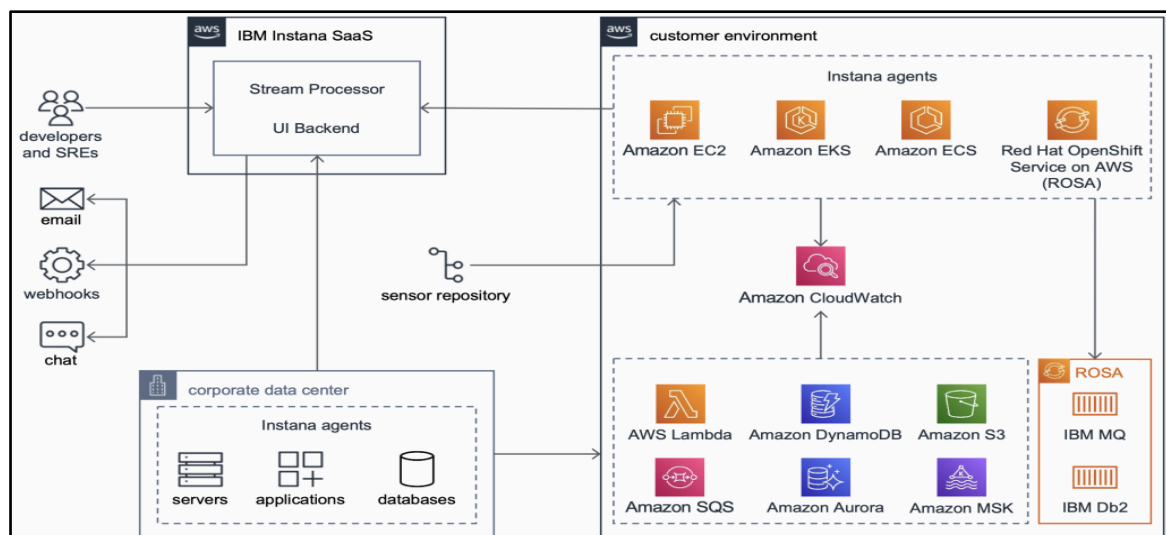
Includes advanced support, billing management, and multiple user access

d) AWS Organization Account

Allows management of multiple AWS accounts under one organization

Useful for companies handling multiple projects or teams

EC2 Architecture Diagram



- **Types of Microsoft Azure Accounts-**

- a) Free Account

- Provides free credits for a limited period + some free services

- Ideal for students and beginners

- b) Pay-As-You-Go Account

- Charges based on actual usage of services

- Flexible and widely used

- c) Enterprise Agreement (EA) Account

- For large businesses with bulk usage

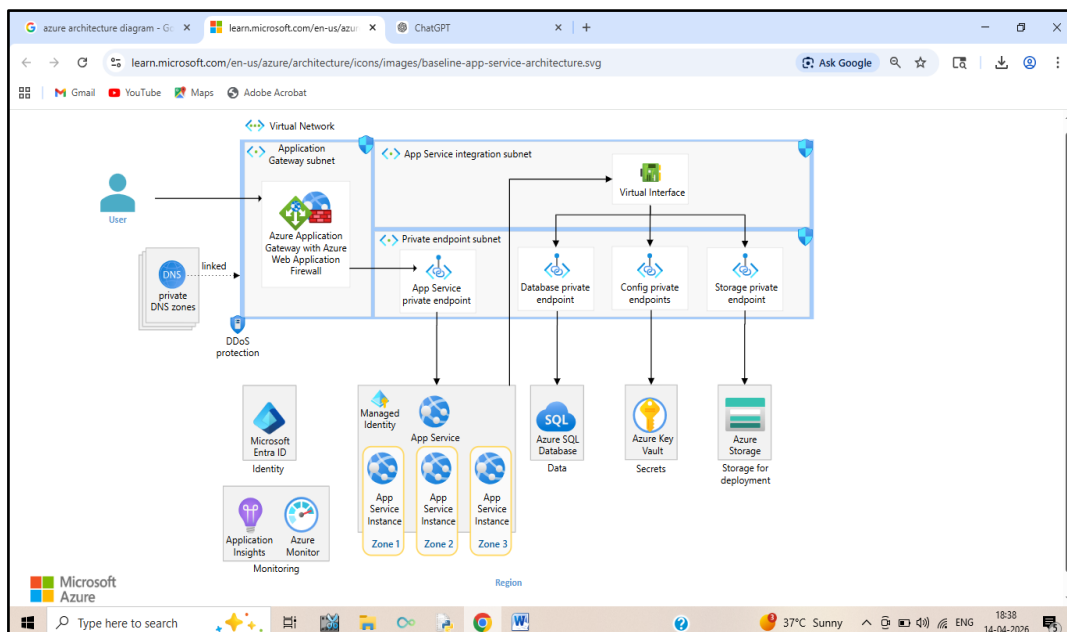
- Offers discounted pricing and centralized billing

- d) Student Account (Azure for Students)

- Free access for students without requiring a credit card

- Includes limited credits and services

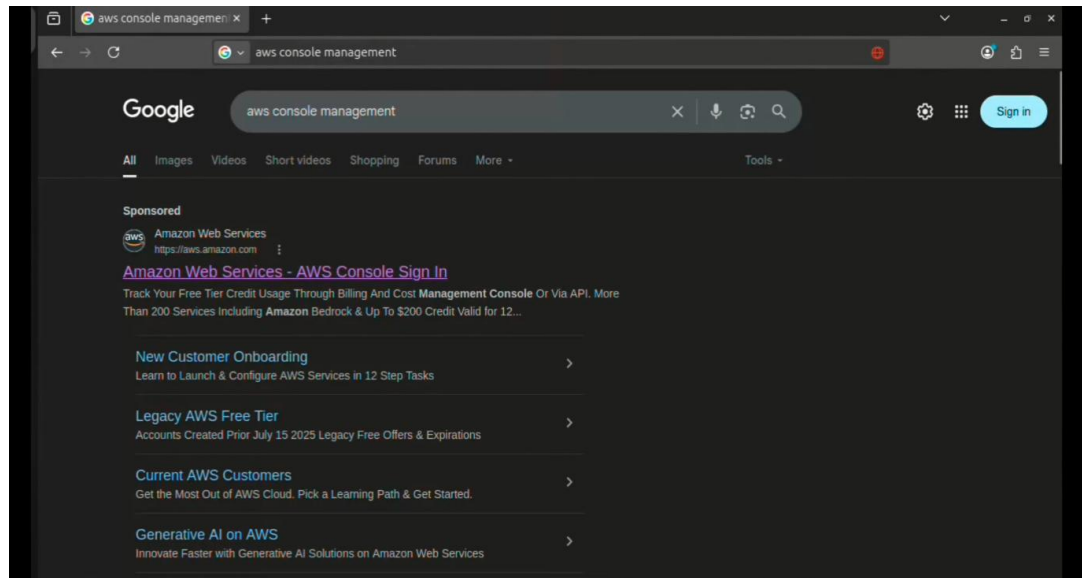
Microsoft Azure Architecture Diagram



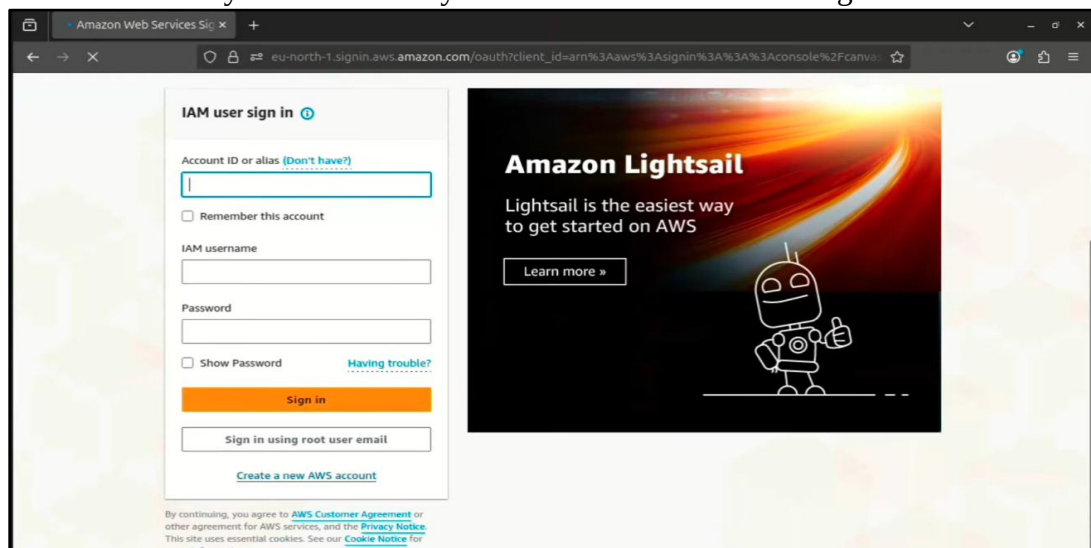
AWS EC2 characteristics: Amazon Web Services EC2 provides on-demand virtual servers that are scalable, flexible, and secure. It follows a pay-as-you-use model where users are charged only for consumed resources. It supports multiple operating systems and can be adjusted based on workload. It is highly available across different regions and data centers.

Azure account characteristics: Microsoft Azure accounts are subscription-based and provide scalable cloud services with global availability. They offer strong security using Microsoft identity management and role-based access control. Resources can be easily managed and organized using resource groups. Azure also supports hybrid cloud integration and works well with Microsoft tools.

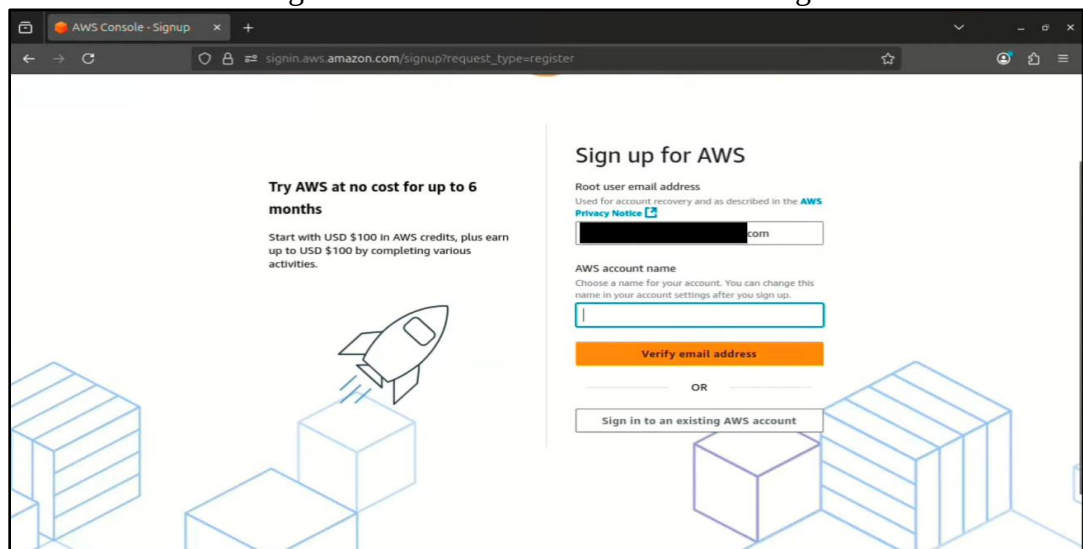
How to Create Amazon AWS Account



If you have already Account then Need not to Register



New Registration for AWS Account with valid gmail ID



Type New Root Password and Confirm Password for Create New Account

The screenshot shows the AWS Console Signup page at `signin.aws.amazon.com/signup?request_type=register`. A modal window titled "Save password for amazon.com?" is open, asking for a username and password. The background page is titled "Sign up for AWS" and "Create your password". It features a rocket launching from blocks. The "Create your password" section has two input fields: "Root user password" and "Confirm root user password", both containing masked text. Below these fields is an orange "Continue (step 1 of 6)" button. An "OR" separator is followed by a link to "Sign in to an existing AWS account".

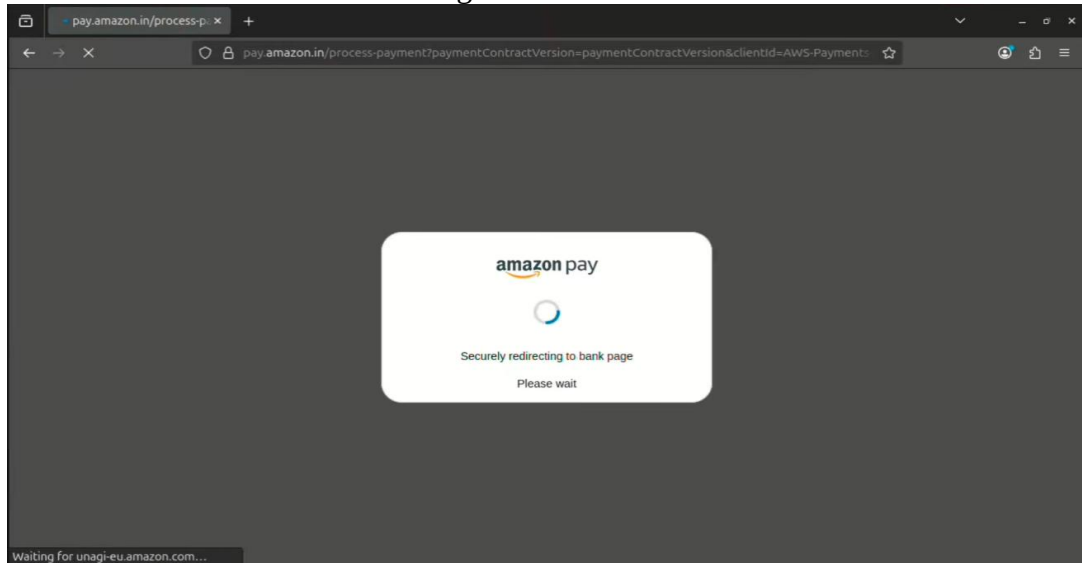
Choose Your Paid or Free Plan for AWS Account

The screenshot shows the AWS Console Signup page at `portal.aws.amazon.com/billing/signup?type=register#/accountplan`. The title is "Choose your account plan". There are two main options: "Free (6 months)" and "Paid". The "Free (6 months)" option includes a list of benefits: "Receive up to \$200 in credits", "Includes free usage of select services", "Workloads scale beyond credit thresholds", and "Access to all AWS services and features". It also notes that after the 6-month free period or when all credits are used, the user can choose to upgrade to a paid plan. The "Paid" option includes similar benefits but adds "Access to all AWS services and features" and "After all of your credits are used, you are charged using pay-as-you-go pricing." Both options have a "Choose free plan" or "Choose paid plan" button. A link to "View additional details" is at the bottom.

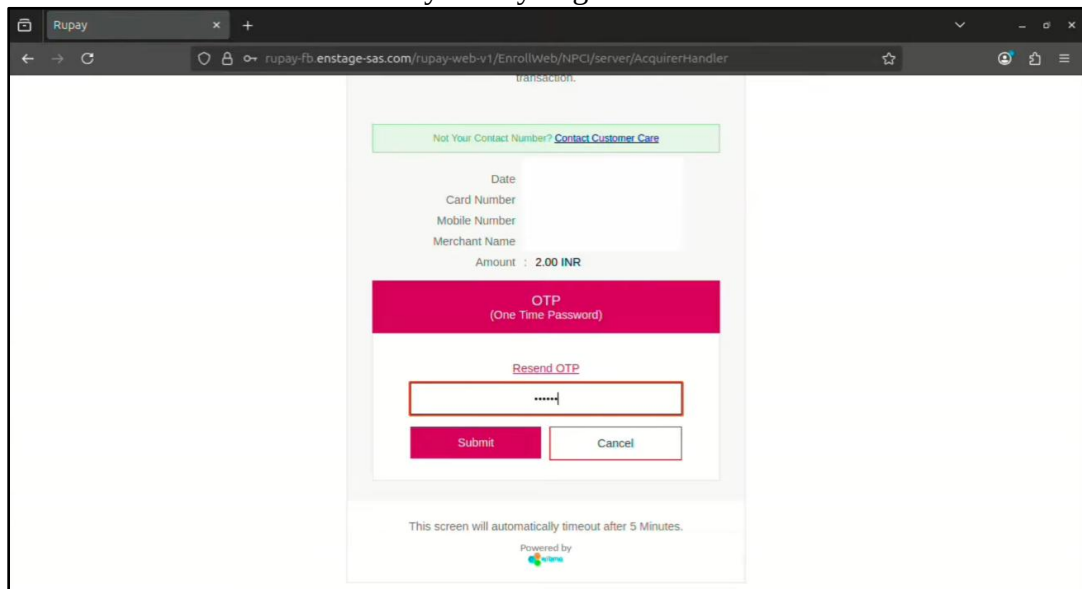
Give Details for AWS Account

The screenshot shows the AWS Console Signup page at `portal.aws.amazon.com/billing/signup?type=register#/account`. The title is "Give Details for AWS Account". On the left, there is a graphic of a dollar bill with the text "additional USD \$100 in credits, such as creating your first AWS budget to monitor cloud costs." On the right, there are several input fields: "Full Name", "Country Code" (set to +1), "Phone Number" (222-333-4444), "Country or Region" (United States), "Address line 1", "Address line 2" (with a hint "Apartment, suite, unit, building, floor, etc."), "City", "State, Province, or Region", and "Postal Code". There are also radio buttons for "Business - for your work, school, or organization" and "Personal - for your own projects", and a question "Who should we contact about this account?".

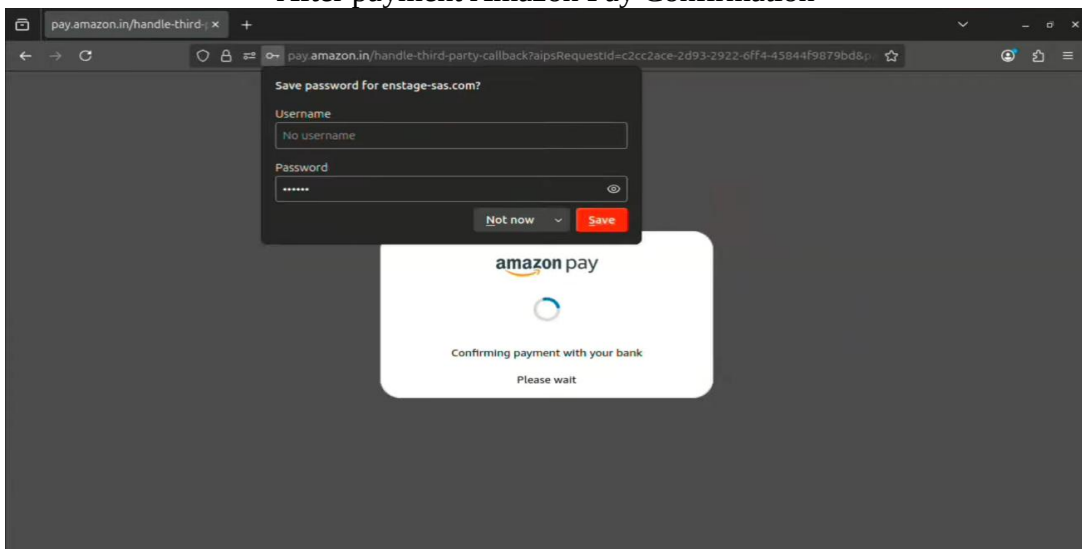
After submitting Card And PAN card details



Verify OTP you get on Email



After payment Amazon Pay Confirmation



Sign up Process for Verify Person

The screenshot shows the 'Sign up for AWS' page with the 'Confirm your identity' section. It includes a 'Primary purpose of account registration' dropdown set to 'Academic', an 'Ownership type' dropdown set to 'Individual', and an 'India document type' dropdown set to 'PAN card'. There are input fields for 'Date of birth' (YYYY/MM/DD) and 'Permanent Account Number (PAN)'. A 'Name' section has a radio button selected for 'Mohseena Maindargi'. At the bottom, there is a 'Choose file' button for uploading a PAN card image.

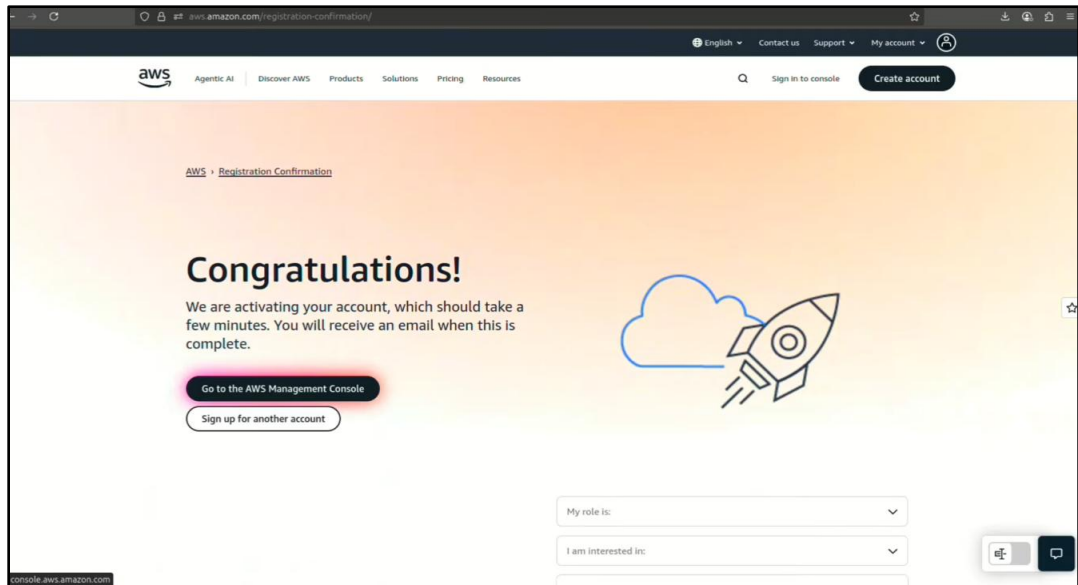
Last step for Confirm Person Identity

The screenshot shows the 'Sign up for AWS' page with the 'Confirm your identity' section. It includes a 'How should we send you the verification code?' section with radio buttons for 'Text message (SMS)' and 'Voice call'. There is a 'Country or region code' dropdown set to 'United States (+1)' and a 'Mobile phone number' input field. A 'Send SMS (step 4 of 5)' button is at the bottom.

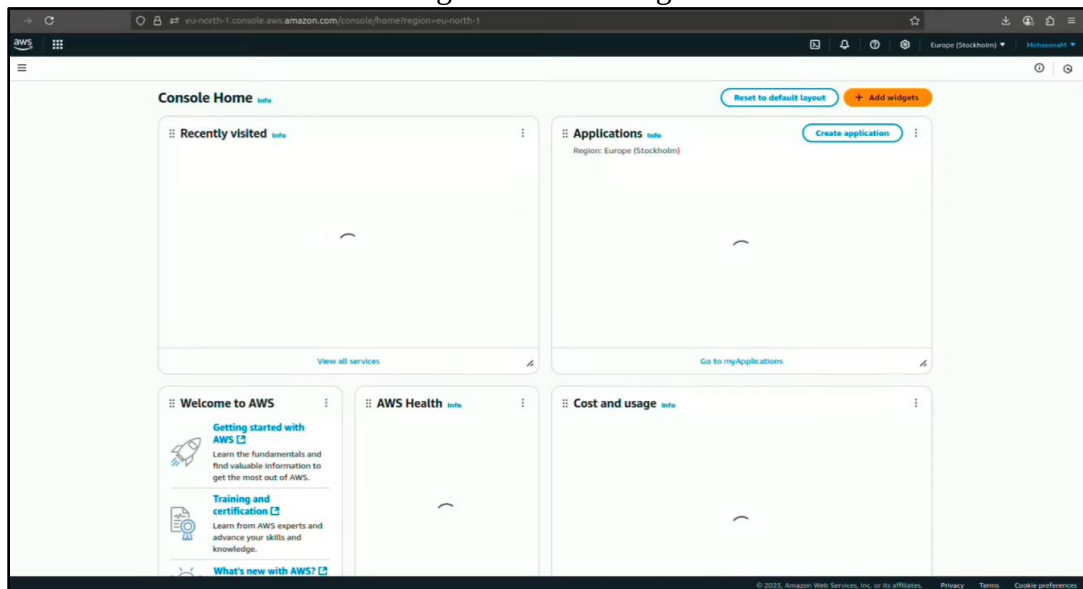
After Submit security code Choose Basic Plan

The screenshot shows the 'Sign up for AWS' page with the 'Select a support plan' section. It includes three plan options: 'Basic support - Free', 'Developer support - From \$29/month', and 'Business support - From \$100/month'. Each plan has a list of features and a corresponding icon. A 'Complete sign up' button is at the bottom.

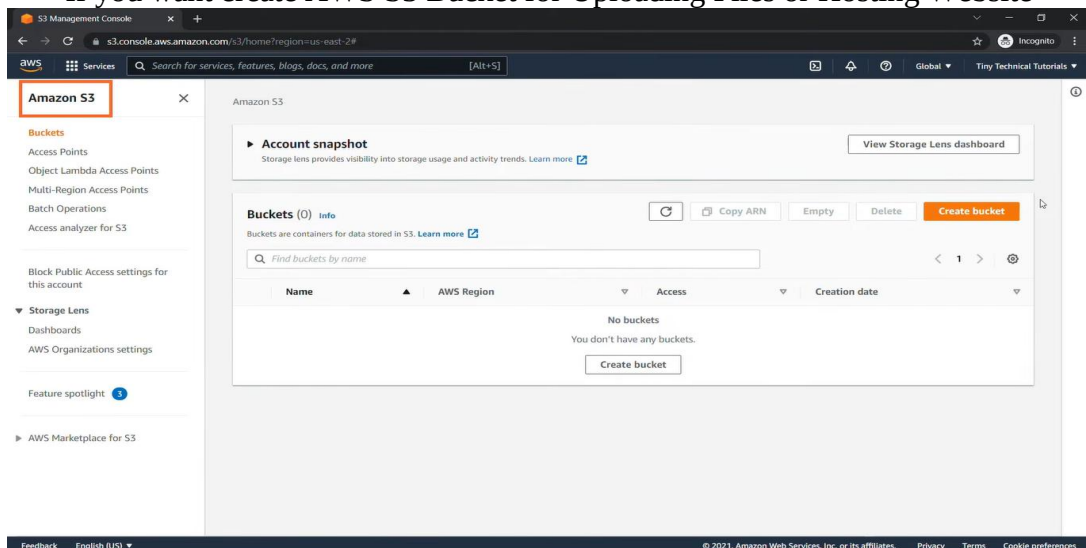
Select AWS Management Console



After Creating AWS Account get Windows



If you want create AWS S3 Bucket for Uploading Files or Hosting Website



Create bucket [Info](#)

Buckets are containers for data stored in S3. [Learn more](#)

General configuration

Bucket name
tinyflix
Bucket name must be unique and must not contain spaces or uppercase letters. [See rules for bucket naming](#)

AWS Region
US East (Ohio) us-east-2

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

Choose bucket

Block Public Access settings for this bucket
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

Public access for this bucket but select private

Block Public Access settings for this bucket
Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☒ **Block all public access**
Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

- ☐ **Block public access to buckets and objects granted through new access control lists (ACLs)**
S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.
- ☐ **Block public access to buckets and objects granted through any access control lists (ACLs)**
S3 will ignore all ACLs that grant public access to buckets and objects.
- ☐ **Block public access to buckets and objects granted through new public bucket or access point policies**
S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.
- ☐ **Block public and cross-account access to buckets and objects through any public bucket or access point policies**
S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Bucket Versioning
Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning
☒ Disable

Select Disable option then On TAGs optional

Tags (1) - optional
Track storage cost or other criteria by tagging your bucket. [Learn more](#)

Key: environment Value - optional: dev Remove

Add tag

Default encryption
Automatically encrypt new objects stored in this bucket. [Learn more](#)

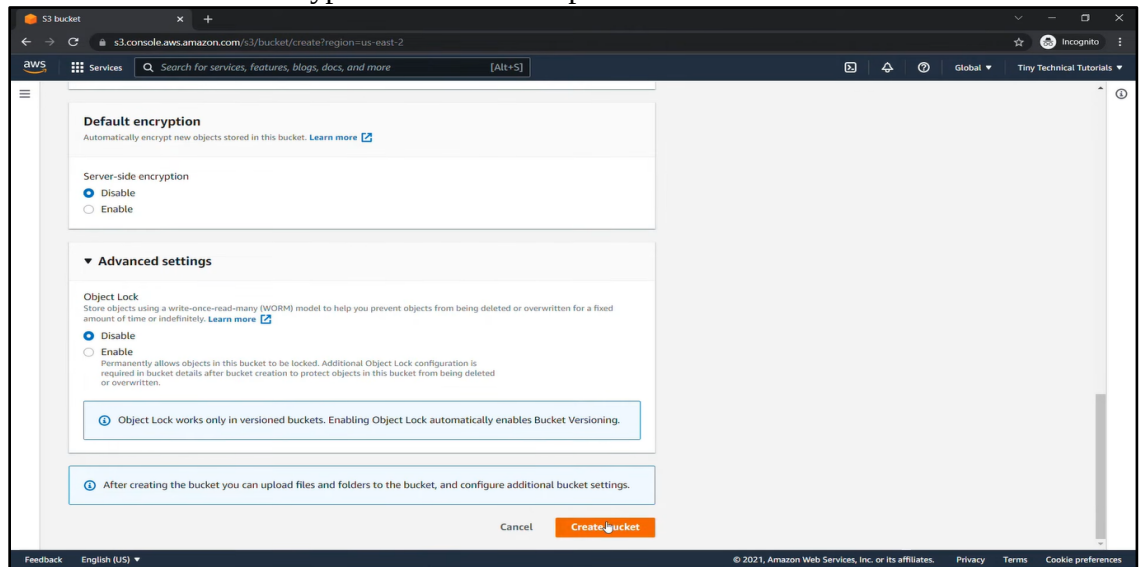
Server-side encryption
☒ Disable
☐ Enable

Advanced settings

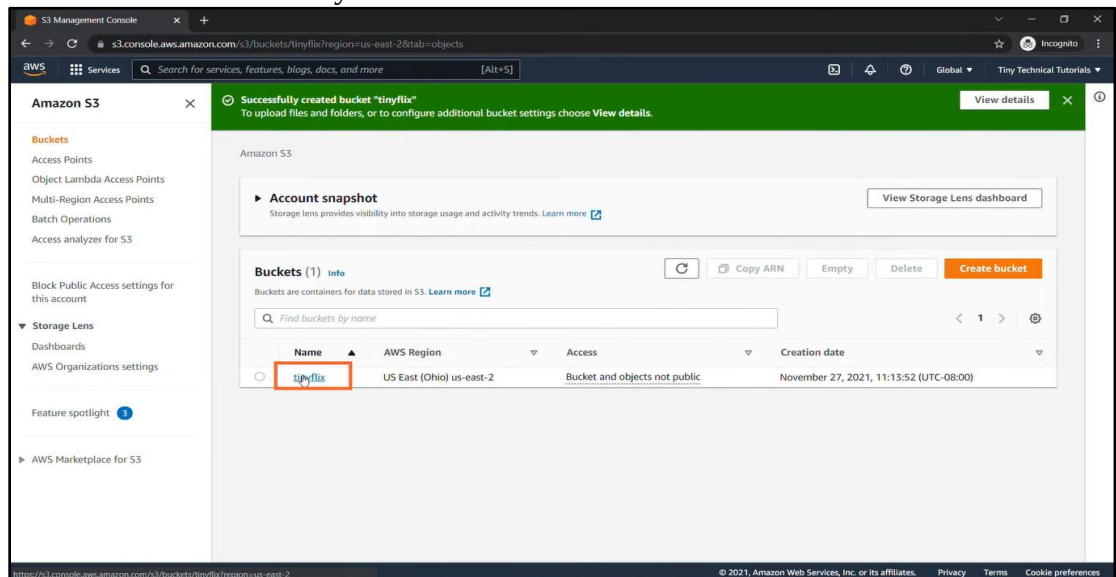
After creating the bucket you can upload files and folders to the bucket, and configure additional bucket settings.

Cancel Create bucket

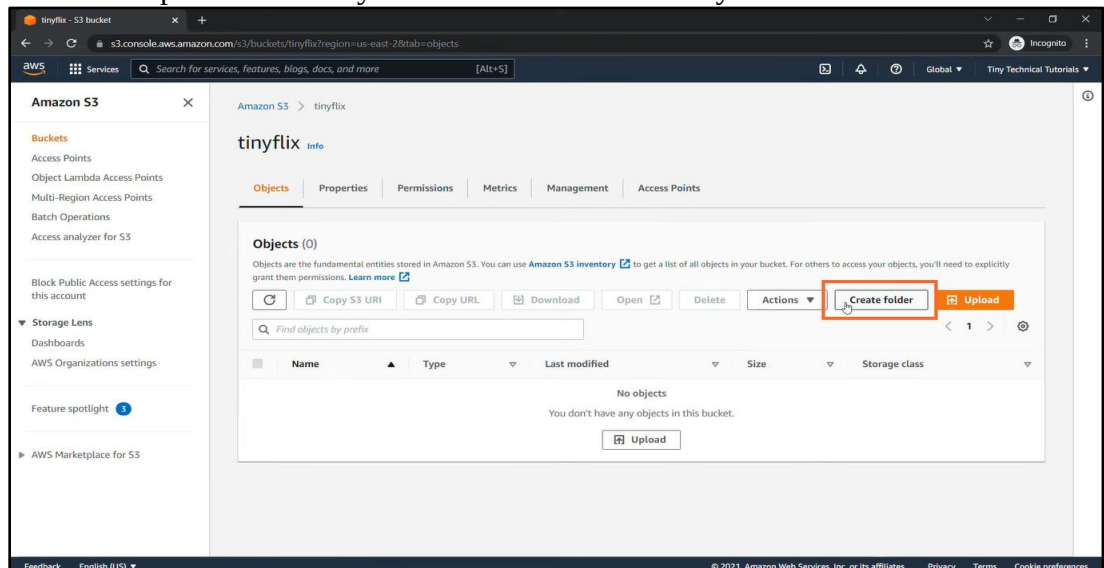
Select Default Encryption tab Disable option then click on create Bucket



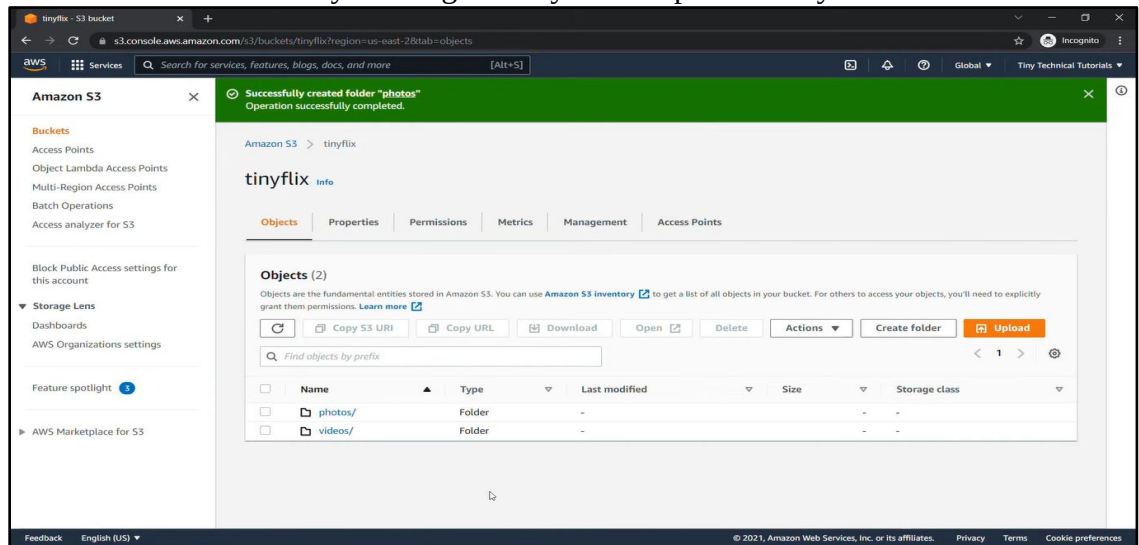
After Successfully Created Bucket. You will see Bucket name.



Upload File first you have to create folder in your Bucket.



After Successfully creating folder you can upload files you have.



Lab Assignment

SET A

1. Create and Host a Static Website Using HTML, CSS on Amazon AWS Account.
2. Install and Create Amazon AWS Account.
3. Create a simple serverless backend application using Azure function.
4. Deploy a Web Application on an Amazon EC2 Virtual Machine.

SET B

1. Create a simple serverless application with API and database into Google Cloud.
2. Install and Create Microsoft Azure Account.
3. Host a Static Website on Amazon S3 Using HTML and CSS.
4. Install and Create Amazon AWS Account.

SET C

1. Create Virtual Machine using Azure Virtual Machine and install Apache Web Server.
2. Create a simple serverless backend application using Azure function.
3. Create and Host a Static Website Using HTML, CSS on Azure Account.
4. Deploy a Webpage on an Amazon EC2 Virtual Server.

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Needs Improvement []

4: Complete []

5: Well Done []

Signature of Instructor:

Date:

Assignment No: 5

Title: Containerization using Docker

Prerequisite:-

- Basic knowledge of using a computer and command line / terminal
- Docker Desktop installed on your machine (Windows / macOS / Linux)
- A text editor (VS Code recommended)
- Active internet connection (to pull images from Docker Hub)
- A web browser (Chrome / Firefox)

Guidelines for Teachers/Instructors:

- Explain the difference between a Virtual Machine and a Container (with a diagram)
- Demonstrate Docker Desktop installation and verify it with `docker run hello-world`
- Explain: Image → Container → Registry workflow with live demo
- Show students how to write a simple Dockerfile and build an image from it
- Introduce Docker Compose with a two-service example before Set C
- Note: All backend code in this assignment uses Python /Flask
- Allocate Set A in slot 1, Set B in slot 2, and Set C in slot 3

Instructions for Students:-

- Install Docker Desktop before the lab session and verify: `docker --version`
- Install Python 3 and verify: `python3 --version`
- Read each question fully before running any command
- Note down every command you use and paste the terminal output in your record
- Take a screenshot of the browser / terminal as proof for each task
- Always stop and remove containers after finishing each question
- Attempt Set A, B, and C only in the slots allocated by the instructor

1. Introduction

Docker is an open-source platform that packages an application together with everything it needs to run — code, libraries, and settings — into a single unit called a container. This guarantees the application behaves the same on every machine, from a student's laptop to a cloud server.

What is Docker?

Docker uses containers — lightweight, isolated processes that share the host operating system's kernel. Unlike virtual machines, containers don't need a full OS each, so they start in seconds and use far less memory.

VM vs. Container — at a glance:

```
Virtual Machine:  Full OS per VM  |  GBs of storage  |  
Minutes to boot  |  High overhead  
Docker Container: Shared OS kernel | MBs of storage  |  
Seconds to start  |  Low overhead
```

Why Use Docker?

- Consistency — "works on my machine" problem is eliminated
- Speed — containers start in seconds
- Portability — run on any OS that supports Docker
- Isolation — each app runs in its own environment, no conflicts
- Easy sharing — push your image to Docker Hub; anyone can pull and run it

2. Core Concepts

2.1 Docker Image

A read-only blueprint for a container, Think of it like a recipe — it describes what software is installed and how the container should be set up. Images are built from a Dockerfile and stored in a registry.

2.2 Docker Container

A running instance of an image. One image can produce many containers. Containers are isolated from each other and from the host machine. When a container is deleted, any data written inside it is lost (unless you use a volume).

2.3 Docker Volume

A way to store data outside the container so it survives container restarts and deletions. Useful for databases and any file a user uploads.

2.4 Docker Network

Containers are isolated by default. Docker networks let containers talk to each other. Containers on the same network can reach each other by name (e.g., container "backend" can call "database" directly).

2.5 Docker Registry / Docker Hub

Docker Hub (hub.docker.com) is the official online store for Docker images. Official images like nginx, mysql, and node are available for free. You can also push your own images to share with others.

3. Essential Docker Commands

Command	What it does
<code>docker --version</code>	Check Docker is installed and show version
<code>docker pull <image></code>	Download an image from Docker Hub
<code>docker images</code>	List all images stored on your machine
<code>docker run <image></code>	Create and start a container from an image
<code>docker run -d <image></code>	Run container in the background (detached)
<code>docker run -p 8080:80 </code>	Map host port 8080 to container port 80
<code>docker run --name x </code>	Give the container a custom name
<code>docker ps</code>	List currently running containers
<code>docker ps -a</code>	List all containers (running + stopped)
<code>docker stop <name/id></code>	Stop a running container
<code>docker rm <name/id></code>	Delete a stopped container
<code>docker rmi <image></code>	Delete a local image
<code>docker logs <name/id></code>	View output / logs of a container
<code>docker exec -it <name> sh</code>	Open a terminal inside a running container
<code>docker build -t myapp .</code>	Build an image from the Dockerfile in this folder
<code>docker compose up -d</code>	Start all services defined in docker-compose.yml
<code>docker compose down</code>	Stop and remove all compose services

4. Writing a Dockerfile

A Dockerfile is a plain text file named exactly Dockerfile (no extension). Docker reads it top-to-bottom to build your image.

Common Instructions

- FROM — choose a base image (always the first line)
- WORKDIR — set the working folder inside the container
- COPY — copy files from your machine into the image
- RUN — run a shell command while building the image (e.g., install packages)
- EXPOSE — tell Docker which port the app listens on
- CMD — the command that runs when the container starts

Dockerfile Template — Static HTML with Nginx

```
# Base image — official Nginx web server
FROM nginx:alpine

# Copy your HTML files into Nginx's web root
COPY . /usr/share/nginx/html

# Nginx listens on port 80 by default
EXPOSE 80
```

Dockerfile Template — Node.js (Express) App

```
FROM node:20-alpine

WORKDIR /app

COPY package*.json ./
RUN npm install --production

COPY . .

EXPOSE 5000
CMD ["node", "server.js"]
```

5. Docker Compose (docker-compose.yml)

Docker Compose lets you define and run multiple containers with a single file. Each container is called a service. Run `docker compose up -d` to start everything at once.

docker-compose.yml Template

```
version: '3.9'

services:

  frontend:          # Service 1 — HTML/JS website
    build: ./frontend
    ports:
      - "3000:80"
    networks:
      - app-net

  backend:           # Service 2 — Flask API
    build: ./backend
    ports:
      - "5000:5000"
    depends_on:
      - database
    environment:
      - DB_HOST=database
    networks:
      - app-net

  database:          # Service 3 — MySQL
```

```
image: mysql:8
environment:
  MYSQL_ROOT_PASSWORD: secret
  MYSQL_DATABASE: myapp
volumes:
  - db-data:/var/lib/mysql
networks:
  - app-net

volumes:
  db-data:

networks:
  app-net:
```

6. How to Dockerize a Python/Flask Project — Step by Step

1. Write your application code (HTML, Python/Flask , etc.)
2. Create a Dockerfile in the project root folder
3. Build the image: `docker build -t myapp:v1 .`
4. Run a container: `docker run -p 8080:5000 myapp:v1`
5. Open your browser at `http://localhost:8080` to verify it works
6. (Optional) Push to Docker Hub: `docker push username/myapp:v1`

7. Practice Programs

Complete these before attempting the formal assignments.

1. Run `docker run hello-world`. Read the output carefully and understand what happened.
2. Pull the nginx image: `docker pull nginx`. Run it on port 8080 and open it in a browser.
3. Run an interactive Alpine Linux container: `docker run -it alpine sh`. Type `ls` and `pwd` inside it.
4. Create a file `hello.html` with the text `<h1>Hello Docker!</h1>`. Run it in an Nginx container.
5. Create a minimal Express app (`server.js`). Dockerize it and verify it responds at `http://localhost:3000`.
6. Build any image, then run `docker images` and `docker ps -a`. Stop and remove the container.

Lab Assignment

Set A

A1. Pull an Image and Explore Docker Commands

Objective: Learn to pull official images from Docker Hub and use basic Docker commands

Steps to follow:

7. Open your terminal. Run `docker --version` to confirm Docker is installed.
8. Pull the official Ubuntu image: `docker pull ubuntu:22.04`.
9. Pull the official Nginx image: `docker pull nginx:latest`.
10. Run `docker images` — note down the Image ID and Size of each image.
11. Run an Nginx container: `docker run -d --name myserver -p 8080:80 nginx`

12. Open `http://localhost:8080` in your browser. Take a screenshot of the Nginx welcome page.
13. Run `docker ps` to see the running container. Note the Container ID and Status.
14. Stop the container: `docker stop myserver`
15. Remove the container: `docker rm myserver`. Verify with `docker ps -a`.

Expected Terminal Output (after `docker ps`):

CONTAINER ID	IMAGE	COMMAND	STATUS
3a7f2c89b1d0	nginx	"/docker-entrypoint..."	Up 2 minutes
		0.0.0.0:8080->80/tcp	myserver

A2. Containerize a Basic HTML Page

Objective:

Build your first custom Docker image that serves a static HTML page.

Create a folder called `mywebsite` with the following two files:

File 1 — `index.html`

```
<!DOCTYPE html>
<html>
<head><title>My Docker Page</title></head>
<body>
  <h1>Hello from Docker!</h1>
  <p>This page is running inside a Docker
container.</p>
  <p>Name: _____ | Roll No: _____</p>
</body>
</html>
```

File 2 — `Dockerfile`

```
FROM nginx:alpine
COPY . /usr/share/nginx/html
EXPOSE 80
```

Steps to follow:

1. Open terminal inside the `mywebsite` folder.
2. Build the image: `docker build -t mywebsite:v1 .`
3. Run a container: `docker run -d --name mysite -p 8081:80 mywebsite:v1`
4. Open `http://localhost:8081` in the browser. Take a screenshot showing your name and roll number.
5. Run `docker images` and record the size of your `mywebsite:v1` image.
6. Stop and remove the container when done.

A3. Explore a Running Container from the Inside

Objective: Learn to open a shell inside a running container and inspect its filesystem

1. Start the Nginx container again: `docker run -d --name explore -p 8082:80 nginx`
2. Open a shell inside the running container: `docker exec -it explore sh`
3. Inside the container, run the following commands one by one and note what you see:
 - `ls /usr/share/nginx/html`
 - `cat /etc/os-release`
 - `hostname`
 - `exit`
4. Back on your machine, run `docker logs explore` to see the Nginx access logs.
5. Run `docker inspect explore` and find the container's IP address in the output.

Set B

B1. Multi-Page Website in a Container

Objective: Containerize a multi-page website with navigation between pages

Create a folder `multisite/` with the following files:

```
multisite/
├── Dockerfile
├── index.html    ← Home page
├── about.html   ← About page
└── contact.html ← Contact page
```

Each HTML page must have a navigation bar linking to all three pages. Use a consistent simple CSS style across all pages. The home page must display your name, roll number, and class.

Steps to follow:

1. Build the image: `docker build -t multisite:v1 .`
2. Run the container: `docker run -d --name mysite -p 8083:80 multisite:v1`
3. Verify all three pages load correctly in the browser and navigation links work.
4. Take a screenshot of each page in the browser.

B2. Two Containers — Password Generator Application

Objective: Build and run a simple 2-container web application using Docker

You will create two containers:

- Container 1 (backend) — a Python/Flask API that generates random passwords
- Container 2 (frontend) — a Nginx container serving an HTML page where the user enters password length and clicks Generate

Folder structure:

```
password-generator/  
├── backend/  
│   ├── Dockerfile  
│   ├── app.py  
│   └── requirements.txt  
└── frontend/  
    ├── Dockerfile  
    └── index.html
```

backend/app.py — Python/Flask API:

```
from flask import Flask, request, jsonify  
from flask_cors import CORS  
import random  
import string  
  
app = Flask(__name__)  
CORS(app)  
  
@app.route('/generate', methods=['GET'])  
def generate_password():  
    length = int(request.args.get('length', 8))  
  
    chars = string.ascii_letters + string.digits +  
    string.punctuation  
    password = ''.join(random.choice(chars) for _ in  
    range(length))  
  
    return jsonify({  
        "length": length,  
        "password": password  
    })  
  
if __name__ == '__main__':  
    app.run(host='0.0.0.0', port=5000)
```

backend/requirements.txt:

```
flask  
flask-cors
```

backend/Dockerfile:

```
FROM python:3.11-slim  
WORKDIR /app  
COPY requirements.txt .  
  
RUN pip install -r requirements.txt  
  
COPY app.py .  
EXPOSE 5000
```



```
CMD ["python", "app.py"]
```

Steps to follow:

1. Build the backend image: `docker build -t backend ./backend`
2. Build the frontend image: `docker build -t frontend ./frontend`
3. Start the backend: `docker run -d --name backend -p 5000:5000 backend`
4. Start the frontend: `docker run -d --name frontend -p 8084:80 frontend`
5. Open `http://localhost:8084` in the browser.
6. Enter a password length and click Generate Password.
7. Verify that a random password is displayed on the webpage.
8. Run `docker ps` to confirm both containers are running.
9. Run `docker logs backend` to verify the backend is serving requests.

Assignment Evaluation

0: Not Done []

1: Incomplete []

2: Late Complete []

3: Needs Improvement []

4: Complete []

5: Well Done []

Signature of Instructor:

Date: